Sensitivity analysis for contamination in egocentric-network randomized trials with interference

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Abstract

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1 Introduction

In causal inference and the design of experiments it is commonly assumed that there is no interference between units, that is, the treatment assigned to one unit does not affect the outcomes of other units (Cox, 1958; Hernán and Robins, 2020). However, researchers are often interested in settings where the treatment of interest can spill over from treated units to untreated ones, leading to interference between units. For example, in infectious disease epidemiology, vaccinating some individuals may provide protection to unvaccinated individuals (Halloran and Struchiner, 1991). The term interference encompasses a wide range of transmission mechanisms, including social influence, information diffusion, contagion, and infectious disease spread. To estimate causal effects under interference, researchers often rely on detailed information about the network structure that encodes the relationships between units. Such data is typically assumed to be on a complete sociocentric network, where a finite population of units and all their connections are observed (Hudgens and Halloran, 2008; Aronow and Samii, 2017; Forastiere et al., 2020; Tchetgen Tchetgen et al., 2020; Ogburn et al., 2022). The finite population can be composed of a collection of disjoint clusters or a single connected network. However, collecting complete sociocentric network data is often logistically challenging and expensive, leading to the use of sampled networks in practice.

A common sampling design is *egocentric network sampling* (Marsden, 2002; Perry et al., 2018), where a set of egos (indexes) are sampled from the population, and each ego recruits a subset of its neighbors in the population network as alters (non-indexes). The observed network is then composed of disjoint ego-networks. This design is attractive due to its logistical feasibility and accessibility. Such egocentric sampling can be combined with randomized experiments, where treatments are randomly assigned to the recruited egos, resulting in an *egocentric-network randomized trial* (ENRT) (Buchanan et al., 2018, 2024; Chao et al., 2025; Fang et al., 2025). ENRTs are increasingly used in various studies, such as HIV prevention (Latkin et al., 2009, 2013; Booth et al., 2016; Miller et al., 2018), substance addiction (Sherman et al., 2009), and mental health (Asmus et al., 2017; Desrosiers et al., 2020).

However, alters may be connected to multiple egos in the population network, and egos may be connected to other egos, resulting in contamination between ego-networks (Torgerson, 2001). Thus, the egocentric sampling process may lead to missing edges in the observed network, leading to misspecification of the interference structure (Weinstein and Nevo, 2023). Previous analyses of ENRTs typically assumed that the ego-networks are *non-overlapping* in the population network (Buchanan et al., 2018, 2024; Chao et al., 2025; Fang et al., 2025). That is, alters are connected only to one ego, and egos are not connected to other egos. Such assumption may not hold in practice and be difficult to verify with the observed data alone.

In this paper, we relax the non-overlapping networks assumption and allow for the possibility of overlapping ego-networks. Under a randomization-based inference paradigm, where the only source of randomness is the treatment assignment, we show that the commonly used Horvitz-Thompson estimators of the indirect and direct effects based on the observed data are biased whenever there is contamination between ego-networks. We develop a sensitivity analysis framework to assess the impact of possible contamination between ego-networks on the causal estimates. We provide bias-corrected estimators of the indirect and direct effects in the presence of contamination between ego-networks. The bias-corrected estimators depend on sensitivity parameters that encode the expected number or the probability of missing alter–ego and ego–ego edges in the population network. These parameters can be specified using covariates and prior knowledge about suspected contamination. For the direct effects on egos, we require an additional sensitivity parameter that relates to the magnitude of the treatment and exposure interaction effect on the egos. We provide practical guidelines on how to specify these parameters in practice, under various scenarios. We propose both a grid sensitivity analysis (Greenland, 1996; Rosenbaum, 2005) and a probabilistic bias analysis (Greenland, 2005; Lash et al., 2014; Fox, 2021) to assess the robustness of the causal estimates to possible contamination. The proposed framework is illustrated through simulations and a real-world application.

The HIV Prevention Trials Network (HPTN) 037 study (Latkin et al., 2009) is used as a motivating example throughout the paper. HPTN 037 is an ENRT that aimed to evaluate the efficacy of a peer education intervention to reduce HIV risk behaviors among people who injects drugs. In it, treated egos were trained on prevention of risky behaviors and encouraged to share this information with their alters. We re-analyze the HPTN 037 study, assessing the robustness of the causal estimates to possible contamination between ego-networks.

Our work relates to several strands of literature. First, it contributes to the growing literature on causal inference under interference where the true network structure is latent and only proxies exists (Toulis and Kao, 2013; Li et al., 2021; Hardy et al., 2019; Boucher and Houndetoungan, 2022; Weinstein and Nevo, 2025). Second, it adds to the literature of interfernce with sampled network data which has primarly focused on random node sampling or snowball sampling designs under linear-in-means parameteric models (Chandrasekhar and Lewis, 2011; Sewell, 2017; Yauck, 2022; Marray, 2024). Third, in contributes to the literature on senstivity analysis under interference, which has mainly focused on unmeasured confounding (VanderWeele et al., 2014), or on settings where offline network is latent and only online network is observed (Egami, 2020). Finally, it relates to the literature on cross-cluster contamination in cluster-randomized trials, which often reccomend changing the clusters boundaries to reduce contamination (Hayes and Moulton, 2017), or more recently, develop data-driven methods for clusters construction (Leung, 2025).

The rest of the paper is organized as follows. In Section 2, we introduce the setup of ENRTs, define the causal estimands of interest, and show that the commonly used estimators based on the observed data are biased under contamination. In Section 3, we develop a sensitivity analysis framework for contamination between ego-networks, providing bias-corrected estimators of the indirect and direct effects. We provide practical guidelines on how to specify the sensitivity parameters in practice. In Section 4, we illustrate the proposed framework via simulations. In Section 5, we re-analyze the HPTN 037 study, assessing the robustness of the causal estimates to possible contamination. Finally, we conclude with a discussion in Section 6. The R package `ENRTsensitivity` that implements the proposed sensitivity analysis method is available at https://github.com/barwein/ENRTsensitivity.

2 Egocentric-network randomized trials

2.1 Setup and recruitment process

Let $\boldsymbol{G} = (\boldsymbol{V}, \boldsymbol{E})$ denote the population network, where $\boldsymbol{V} = \{1, \ldots, N\}$ is the set of units, and $\boldsymbol{E} = \{(i,j) : i,j \in \boldsymbol{V}\}$ is the set of edges, assumed to be binary. The population is of size $N = |\boldsymbol{V}|$, which is assumed to be a finite, unknown number. The population network $\boldsymbol{G}$ can be described by its adjacency matrix $\boldsymbol{A}$, where $A_{ij} = 1$ if edge $(i,j) \in E$, and 0 otherwise. We

assume that the population network is undirected and and that there are no self-edges, that is, $A_{ij} = A_{ji}$ and $A_{ii} = 0$ for all i.

Units are recruited to the study via an *egocentric sampling design*. The sampling process consists of two stages. In the first stage, a set of egos (indexes) are sampled from the population. Then, in the second stage, each ego recruits a subset of its neighbors in the population network as alters (non-indexes). Formally, the recruitment process is as follows:

1. Sample egos from the population. Denote the set of recruited egos by $\mathcal{R}_e \subset \boldsymbol{V}$, and $n_e = |\mathcal{R}_e|$ as the number of egos. We do not impose any restrictions on the sampling process of egos.

2. Each ego recruits alters from its neighbors in the population network that are not recruited as egos in the first stage. We assume that each alter is recruited by a single ego, resulting in n_e disjoint ego-networks. This is often enforced in the recruitment process by researchers to avoid overlaps between the observed ego-networks. Denote the set of recruited alters by $\mathcal{R}_a \subset \boldsymbol{V} \setminus \mathcal{R}_e$, and let $n_a = |\mathcal{R}_a|$ be their number.

The set of all recruited units is $\mathcal{R} = \mathcal{R}_e \cup \mathcal{R}_a$ resulting in an egocentric sample of size $n = n_e + n_a$, which is assumed to be a strict subset of the population $n \ll N$. The observed network $\widetilde{\boldsymbol{G}} = (\mathcal{R}, \widetilde{\boldsymbol{E}})$ is a subgraph of the population network $\boldsymbol{G}$ composed of n_e disjoint ego-networks. The set of observed edges is $\widetilde{\boldsymbol{E}} = \cup_{i \in \mathcal{R}_e} \{(i,j) : e(j) = i\}$, where $e(j) \in \mathcal{R}_e$ is the ego of alter $j \in \mathcal{R}_a$. Clearly, $\widetilde{\boldsymbol{E}} \subseteq \boldsymbol{E}$. Let $\widetilde{\boldsymbol{A}}$ be the adjacency matrix of the observed network $\widetilde{\boldsymbol{G}}$.

Figure 1 illustrates the egocentric sampling process in a small population. Although the sampled egos are connected and some alters belong to multiple ego-networks in the population network, each ego reports only a subset of its true neighborhood, yielding an observed network made up of disjoint ego-networks.

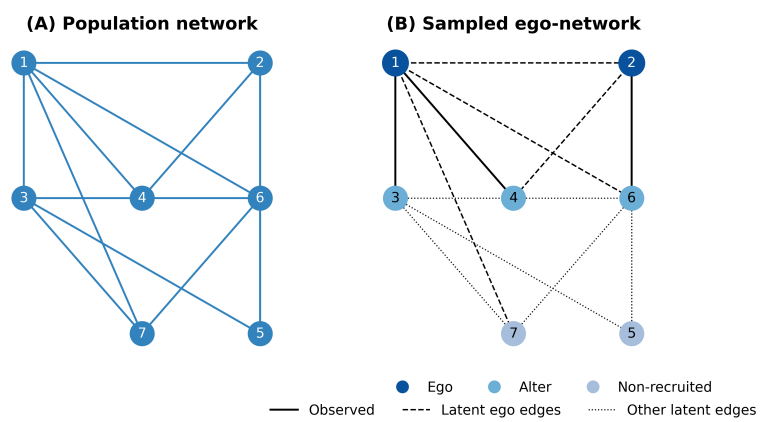

Figure 1: Egocentric sampling. Panel (A) shows the full $N = 7$ population network. Panel (B) illustrates an egocentric sample: each ego (dark blue, $n_e = 2$) reports only a subset of its true neighborhood as alters (medium blue, $n_a = 3$). The observed network is composed of disjoint ego–networks stars (solid lines), while ego–ego and unreported ego–alter connections remain latent (dashed lines). All other edges are also unobserved (dotted lines).

2.2 Experimental design, potential outcomes, and causal estimands

In what follows, all assumptions, estimands, and estimators are conditioned on the recruited subsample. That is, we consider in-sample causal effects. Let $\boldsymbol{Z}$ be the binary treatment assignment vector, where $Z_i = 1$ if unit i is treated, and 0 otherwise. Assume that treatments are randomly assigned to the recruited egos via Bernoulli randomization.

Assumption 1 (Bernoulli experimental design)**.** *Treatments are independently assigned only to recruited egos* $\Pr(Z_i = 1 \mid \mathcal{R}) = p_z \mathbb{I}\{i \in \mathcal{R}_e\}$ *where* $p_z \in (0,1)$ *is a known constant.*

Denote the treatment space as $\mathcal{Z} = \left\{\boldsymbol{z} \in \{0,1\}^N : \Pr(\boldsymbol{Z} = \boldsymbol{z} \mid \mathcal{R}) > 0\right\}$, where treatments of non-recruited units are assumed to be equal to zero with probability one. This experimental design, in which treatments are randomized to egos following an egocentric sample, is referred to as an *egocentric-network randomized trial* (ENRT).

Let $Y_i(\boldsymbol{z})$ be the potential outcomes of each unit under treatment $\boldsymbol{z} \in \mathcal{Z}$. We take a randomization-based inference approach, where the potential outcomes assumed to be fixed and the only source of randomness is the treatment assignment. Let $\boldsymbol{Z}_{-i}$ denote the treatment assignment of all units except i. Assume that $\boldsymbol{Z}_{-i}$ affects the potential outcomes of unit i only through the values of the following exposure mapping (Manski, 2013; Aronow and Samii, 2017; Forastiere et al., 2020).

Assumption 2 (Exposure mapping)**.** *Let* $F(\boldsymbol{Z}_{-i}, \boldsymbol{A}) = \mathbb{I}\left\{\sum_{j \neq i} Z_j A_{ij} > 0\right\}$. *For any* $\boldsymbol{z}, \boldsymbol{z}' \in \mathcal{Z}$, *if* $z_i = z_i'$ *and* $F(\boldsymbol{z}_{-i}, \boldsymbol{A}) = F(\boldsymbol{z}'_{-i}, \boldsymbol{A})$, *then* $Y_i(\boldsymbol{z}) = Y_i(\boldsymbol{z}')$.

Denote the exposures in the true population network by $F_i = F(\boldsymbol{Z}_{-i}, \boldsymbol{A})$. The exposures F_i indicate whether at least one of the neighbors of unit i is treated. Assumption 2 implies that the exposure mapping is correctly specified (Sävje, 2023), and that only treatments of neighbors affect the potential outcomes of unit i. This binary exposure mapping specification is common in the ENRT literature (Buchanan et al., 2018; Chao et al., 2025; Fang et al., 2025). By Assumption 2, we can express the potential outcomes as $Y_i(\boldsymbol{Z}) = Y_i(Z_i, F_i)$. Consequently, each unit has four effective potential outcomes, $Y_i(z, f)$ for $z, f \in \{0,1\}$, indicating whether unit i is treated and whether at least one of its neighbors is treated. We assume that the observed outcomes Y_i, $i \in \mathcal{R}$ are generated from one of the potential outcomes by:

Assumption 3 (Consistency)**.** $Y_i = \sum_{z,f \in \{0,1\}} Y_i(z,f)\mathbb{I}\{Z_i = z, F_i = f\}$ *for all* $i \in \mathcal{R}$.

In addition, assume that we observe covariates $\boldsymbol{X}_i$ for each recruited unit.

We consider two types of causal estimands, separate for egos and alters. The first is the sample average indirect effects of treatment on the alters

$$IE = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} Y_i(0,1) - Y_i(0,0). \tag{1}$$

Eq. (1) represents the average effect on alters of being exposed to a treated neighbor compared to not being exposed. In Appendix A, we also consider the indirect effects on the relative risk scale, in the case of binary outcomes. The second estimand is the sample average direct effect of treatment on the egos

$$DE = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} Y_i(1,0) - Y_i(0,0). \tag{2}$$

Eq. (2) represents the average effect on egos of being treated compared to not being treated when not exposed to any treated neighbors. We explicitly separate the causal estimands for egos and alters. This separation resolve interpretation issues of other estimands (Crawford et al., 2019). Furthermore, by not conflating the two groups we avoid imposing additional assumptions and structure, as the recruitment process (Section 2.1) may induce different selection mechanisms for egos and alters, leading to incomparable distributions of potential outcomes between the two groups.

2.3 Estimation with the observed data

Researchers have access only to the observed data composed of the treatment assignments $\boldsymbol{Z}$, the observed outcomes Y_i, the observed network $\widetilde{\boldsymbol{A}}$, and possibly covariates $\boldsymbol{X}_i$. By Assumption 3, the observed outcomes Y_i are generated from the potential outcomes with exposures F_i computed using the *population network*. The observed exposures are $\widetilde{F}_i = F(\boldsymbol{Z}_{-i}, \widetilde{\boldsymbol{A}})$, which may differ from the true exposures F_i. That is, the observed network $\widetilde{\boldsymbol{A}}$ might not include all the relevant edges present in the population network $\boldsymbol{A}$ due to the egocentric sampling process, resulting in *misspecified network interference structure* (Weinstein and Nevo, 2023). We specifically care about missing ego–ego and alter–ego edges induced by the sampling process.

Consider, for example, the context of Figure 1. If only ego 1 is treated, for alter 6 we observe exposure $\widetilde{F}_i = 0$ since its ego (node 2) is not treated, but its true exposure is $F_i = 1$ due to its link to ego 1 in the population network. Similarly, for ego 2 we observe exposure $\widetilde{F}_i = 0$ while $F_i = 1$ in the population network since it is connected to ego 1.

By the ENRT design, the observed exposures of egos are $\widetilde{F}_i = 0$ for all $i \in \mathcal{R}_e$, while the observed exposures of alters can be either zero or one, depending on their ego's treatment status. Specifically, by Assumption 1, alters are exposed to a treated ego in the observed network with probability $\Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) = p_z$.

Researchers can estimate causal effect only with the observed exposures. The most common estimator in randomization-based inference settings is the Horvitz-Thompson (HT) estimator. For the indirect effects (1), its form is

$$\widehat{IE} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right], \tag{3}$$

and for direct effects (2),

$$\widehat{DE} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \left[\frac{\mathbb{I}\{Z_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{Z_i = 0\} Y_i}{1 - p_z} \right]. \tag{4}$$

If some alter–ego or ego–ego edges are missing in the observed network $\widetilde{\boldsymbol{A}}$ due to the egocentric sampling process, then for some units the observed exposures $\widetilde{F}_i$ may not agree with the true exposures F_i under all possible treatment assignments $\mathcal{Z}$. In that case, both estimators (3) and (4) may not be design-unbiased.

Proposition 1. *Under Assumptions 1, 2, 3,*

$$\mathbb{E}_{\boldsymbol{Z}} \left[\widehat{IE} \right] = IE + \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{p_z - \pi_i^a}{1 - p_z} \left[Y_i(0,1) - Y_i(0,0) \right],$$

$$\mathbb{E}_{\boldsymbol{Z}} \left[\widehat{DE} \right] = DE + \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \pi_i^e \left[\{Y_i(1,1) - Y_i(0,1)\} - \{Y_i(1,0) - Y_i(0,0)\} \right],$$

where $\pi_i^a = \Pr(F_i = 1 \mid i \in \mathcal{R}_a)$ and $\pi_i^e = \Pr(F_i = 1 \mid i \in \mathcal{R}_e)$ are the probabilities that an alter and ego, respectively, are exposed to at least one treated neighbor in the population network $\boldsymbol{A}$.

When an alter is connected only to its own ego, then $\pi_i^a = p_z$, and (3) is a design-unbiased estimator of (1). Otherwise, whenever an alter is connected to more than one ego in the population network, $\pi_i^a > p_z$, and the indirect effect estimator (3) based on the observed exposures is biased. The magnitude of the bias depends on the number of egos each alter is connected to in the population network. That is, the more egos an alter is connected to, the larger the gap between π_i^a and p_z, and the larger the bias. In addition, the naive indirect estimator (3) is attenuated towards the null of no indirect effect (Appendix A).

If an ego is not connected to any other egos in the population network, then $\pi_i^e = 0$. In that case, (4) is a design-unbiased estimator of (2). However, if an ego is connected to at least one other ego in the population network, then $\pi_i^e > 0$, and the direct effect estimator (4) based on the observed data is biased. Similarly to alters, the magnitude of the bias depend on the number ego–ego edges to in the population network.

3 Sensitivity analysis for contamination

Proposition 1 shows that the HT estimators based on the observed data are biased whenever there is contamination between the ego-networks. We refer to contamination as the presence alter–ego and ego–ego edges in the population network $\boldsymbol{A}$, that are missing from the observed network $\widetilde{\boldsymbol{A}}$ due to the egocentric sampling process. Under such contamination, the ego-networks are not disjoint in the population network, leading to incorrect exposures when computed using the observed network.

By the structure of the egocentric sampling and the exposure mapping specification (Assumption 2), alters with observed exposure $\widetilde{F}_i = 1$ will be correctly classified. That is $\widetilde{F}_i = 1$ implies $F_i = 1$ with probability one for $i \in \mathcal{R}_a$. However, alters with $\widetilde{F}_i = 0$ can have a true exposure of $F_i = 1$ or $F_i = 0$, depending on the structure of the population network $\boldsymbol{A}$. Therefore, the observed outcomes Y_i for alters with $\widetilde{F}_i = 0$ can be generated from $Y_i(0,1)$ or $Y_i(0,0)$.

On the other hand, all egos have observed exposures of $\widetilde{F}_i = 0$, as they are linked only to alters in their ego-network in the observed network $\widetilde{\boldsymbol{A}}$. However, the observed outcomes of egos that are assigned to treatment $Z_i = z$ can be generated from $Y_i(z,1)$ or $Y_i(z,0)$, implying that observed outcomes of egos are possibly linked to four different potential outcomes, compared to only two among the alters.

We develop a sensitivity analysis framework that enables researchers to assess the impact of possible contamination between ego-networks on the causal estimates. The framework provides bias-corrected estimators of the indirect and direct effects that account for possible exposure misspecification due to the egocentric-network recruitment process.

The bias-corrected indirect and direct effect estimators are a function of the exposure probabilities under the true population network π_i^a and π_i^e, respectively. The structure of the egocentric sampling design and our setup enables us to decompose π_i^a and π_i^e into basic components that are straightforward to compute given specification of the *expected number* or the *probability* of missing alter–ego and ego–ego edges. The latter will form the basis of our sensitivity parameters, enabling seamless integration of covariates and prior knowledge into the sensitivity analysis. We provide multiple examples and practical guidelines on how to specify these parameters in Section 3.3.

For the direct effects, we require an additional sensitivity parameter to account for the complexity resulting from possibly four potential outcomes generating the observed ones. This parameter is related to the magnitude of the treatment and exposure interaction effect on the egos, and its specifics are discussed in Section 3.2.

3.1 Indirect effects

Define the bias-corrected indirect effect estimator as a function of the observed data and π_i^a

$$\widehat{IE}_{adj} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{1-p_z}{1-\pi_i^a} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1-p_z} \right], \tag{5}$$

Given π_i^a, (5) is a design-unbiased estimator of (1)

Proposition 2. *Under Assumptions 1, 2, 3, we have* $\mathbb{E}_{\boldsymbol{Z}} \left[\widehat{IE}_{adj} \right] = IE$.

By the ENRT design and Assumptions 2, $\pi_i^a \geq p_z$, with equality if and only if there are no additional latent alter–ego edges. Consequently, if each alter has the same exposure probabilities $\pi_i^a = \pi^a$, then $\widehat{IE}_{adj} = \frac{1-p_z}{1-\pi^a} \widehat{IE}$. Therefore, $\text{sign}\left(\widehat{IE}_{adj}\right) = \text{sign}\left(\widehat{IE}\right)$, and $\widehat{IE}_{adj} \geq \widehat{IE}$ if $\widehat{IE} > 0$ and vice versa otherwise.

The bias-corrected estimator (5) can be modified to include an outcome model. Let $\mu_i^a(f) = \mathbb{E}\left[Y_i \mid \widetilde{F}_i = f, \boldsymbol{X}_i, i \in \mathcal{R}_a \right]$ be the conditional expectation among alters with observed exposure $\widetilde{F}_i = f$ and covariates $\boldsymbol{X}_i$. The augmented bias-corrected estimator is

$$\begin{aligned} \widehat{IE}_{adj}^{aug} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \frac{1-p_z}{1-\pi_i^a} \Bigg[& \frac{\mathbb{I}\{\widetilde{F}_i = 1\} \left(Y_i - \mu_i^a(1)\right)}{p_z} - \\ & \frac{\mathbb{I}\{\widetilde{F}_i = 0\} \left(Y_i - \mu_i^a(0)\right)}{1-p_z} + \{\mu_i^a(1) - \mu_i^a(0)\} \Bigg]. \end{aligned} \tag{6}$$

Given that $\mu_i^a(f)$ are estimated in a way that ensure their independence from the treatment assignment $\boldsymbol{Z}$, e.g., via cross-fitting (Chernozhukov et al., 2018), then (6) is also a design-unbiased estimator of (1), but can have lower variance.

Given π_i^a, the variance of the bias-corrected estimators (5) and (6) can be estimated using standard techniques for HT estimators under Bernoulli designs or thorough a nonparameteric bootstrap that resamples ego-networks (see Appendix A for details).

3.2 Direct effects

As the observed outcomes of egos are possibly generated from four different potential outcomes, compared to only two for the alters, bias correction for the direct effect estimator requires an additional structural assumption. Specifically, we require that the unit-level direct effects among exposed ($F_i = 1$) and non-exposed ($F_i = 0$) egos are proportional.

Assumption 4. *There exists a constant κ such that $Y_i(1,1) - Y_i(0,1) = \kappa \{Y_i(1,0) - Y_i(0,0)\}$ for all egos $i \in \mathcal{R}_e$.*

The interpretation of the additional sensitivity parameter κ in Assumption 4 is related to the interaction term between treatment and exposure in the potential outcome model of the egos. Consider the saturated potential outcomes model

$$Y_i(z, f) = \beta_{0i} + \beta_{1i} z + \beta_{2i} f + \beta_{3i} z f,$$

then Assumption 4 is equivalent to $\beta_{3i} = (\kappa - 1)\beta_{1i}$. That is, the interaction term β_{3i} is proportional to the direct effect term β_{1i} for all egos. Reasoning about the range of plausible κ values depends on the mechanism under study. For example, in vaccine efficacy studies, if treatment represents vaccines, it is reasonable to assume that direct effects among egos with only non-vaccinated neighbors will be larger than those among egos exposed to vaccinated units, and therefore $\kappa < 1$. On the other hand, in the HPTN-037 study (Latkin et al., 2009), which we analyze in Section 5, treatment represent a behavioral intervention. Such intervention is likely to have monotonic effects, namely, the direct effect will be larger for egos exposed to a treated neighbor, leading to $\kappa > 1$.

Define the the bias-corrected direct effect estimator as a function of the observed data, π_i^e, and κ

$$\widehat{DE}_{adj} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \frac{1}{1 + \pi_i^e(\kappa - 1)} \left[\frac{\mathbb{I}\{Z_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{Z_i = 0\} Y_i}{1 - p_z} \right] \tag{7}$$

Given π_i^e and κ, (7) is a design-unbiased estimator of (2)

Proposition 3. *Under Assumptions 1, 2, 3, 4, we have $\mathbb{E}_{\boldsymbol{Z}} \left[\widehat{DE}_{adj} \right] = DE$.*

If $\pi_i^e = 0$ or $\kappa = 1$, the bias-corrected estimator will reduce to the naive HT estimator (4), and both will be unbiased. Namely, the naive estimator (4) is unbiased if there are no latent ego–ego edges or if the interaction term between treatment and exposure is zero.

Moreover, under Assumption 4, the naive estimator $\widehat{IE}$ is attenuated *away* from the null if $\kappa > 1$, and attenuated *towards* the null if $\kappa \in \left(\min_{i \in \mathcal{R}_e} \left\{ 1 - \frac{1}{\pi_i^e} \right\}, 1 \right)$. See Appendix A for details. However, the left bound is not informative for most plausible κ values. For example, if $\pi_i^e = \pi^e = 0.01$, the the lower bound is -99, which is not practically meaningful. Thus, in general, the naive estimator is attenuated towards the null if $\kappa < 1$. In addition, for most plausible κ values $\text{sign}\left(\widehat{DE}_{adj}\right) = \text{sign}\left(\widehat{DE}\right)$, so the bias-corrected estimator preserves the direction of the naive estimator.

Similarly to the augmented indirect effect estimator (6), the bias-corrected direct effect estimator (7) can also be extended to include an outcome model. Let $\mu_i^e(z) = \mathbb{E}\left[Y_i \mid Z_i = z, \boldsymbol{X}_i, i \in \mathcal{R}_e\right]$. The augmented bias-corrected estimator is

$$\begin{aligned} \widehat{DE}_{adj}^{aug} = \frac{1}{n_e} \sum_{i \in \mathcal{R}_e} \frac{1}{1 + \pi_i^e(\kappa - 1)} \Bigg[& \frac{\mathbb{I}\{Z_i = 1\} \left(Y_i - \mu_i^e(1)\right)}{p_z} - \\ & \frac{\mathbb{I}\{Z_i = 0\} \left(Y_i - \mu_i^e(0)\right)}{1 - p_z} + \{\mu_i^e(1) - \mu_i^e(0)\} \Bigg]. \end{aligned} \tag{8}$$

3.3 Specification of the sensitivity parameters

Propositions 2-3 establish that the bias-corrected estimators of the indirect and direct effects depend on the exposure probabilities $\pi_i^a = \Pr(F_i = 1 \mid i \in \mathcal{R}_a)$ and $\pi_i^e = \Pr(F_i = 1 \mid i \in \mathcal{R}_e)$ of alters and egos in the population network $\boldsymbol{A}$. For the direct effects, the bias-corrected estimator also depends on the sensitivity parameter κ. We now illustrate how the computation of π_i^a and π_i^e reduce to specification of edge-level probabilities or the expected number of missing alter–ego and ego–ego edges. The latter form the basis of our sensitivity parameters.

By the exposure mapping specification (Assumption 2) and the experimental design (Assumption 1), the probabilities π_i^a and π_i^e depend only on alter–ego or ego–ego edges in the population network $\boldsymbol{A}$. In the observed network $\widetilde{\boldsymbol{A}}$, each alter $i \in \mathcal{R}_a$ is connected only to its ego $e(i) \in \mathcal{R}_e$. Let $\mathcal{R}_e^j = \mathcal{R}_e \setminus \{j\}$ be the set of all egos other than $j \in \mathcal{R}_e$. The latent alter–ego edges $\left\{A_{ij} : i \in \mathcal{R}_a, j \in \mathcal{R}_e^{e(i)}\right\}$, and ego–ego edges $\{A_{ij} : i \neq j \in \mathcal{R}_e\}$ are the missing edges from $\widetilde{\boldsymbol{A}}$ relevant to the exposure probabilities. Define the edge-level probabilities by

$$\begin{aligned} &\Pr\left(A_{ij} = 1 \mid i \neq j \in \mathcal{R}_e\right) = \rho_{ij}^e, \\ &\Pr\left(A_{ij} = 1 \mid i \in \mathcal{R}_a, j \in \mathcal{R}_e^{e(i)}\right) = \rho_{ij}^a, \end{aligned} \tag{9}$$

for $\rho_{ij}^e, \rho_{ij}^a \in [0, 1]$. Using (9), we can derive expressions for π_i^e and π_i^a. By the law of total probability and Assumption 1, $\Pr\left(Z_j A_{ij} = 1 \mid i \neq j \in \mathcal{R}_e\right) = p_z \rho_{ij}^e$. Assuming (conditional) edge independence yields,

$$\pi_i^e = 1 - \prod_{j \in \mathcal{R}_e^i} \left(1 - p_z \rho_{ij}^e\right). \tag{10}$$

Alter $i \in \mathcal{R}_a$ is exposed ($F_i = 1$) only if its ego $e(i)$ or at least one of the other recruited egos connected to i in the population network $\boldsymbol{A}$ are treated. Therefore, we obtain (Appendix A),

$$\pi_i^a = p_z + (1 - p_z) \left[1 - \prod_{j \in \mathcal{R}_e^{e(i)}} \left(1 - p_z \rho_{ij}^a\right)\right]. \tag{11}$$

Eqs. (10) and (11) illustrate how the exposure probabilities can be computed given specifications of the edge-level probabilities ρ_{ij}^e and ρ_{ij}^a. The latter form the basis of our sensitivity parameters. We now consider some prominent examples of how researchers can specify these parameters in practice. The examples include both homogeneous and heterogeneous edge probabilities, as well as specifications based on the expected number of missing edges.

Example 1 (Homogeneous probabilities)**.** Specify the same probabilities (9) for all units: $\rho_{ij}^e = \rho^e$ and $\rho_{ij}^a = \rho^a$. The exposure probabilities (10) and (11) simplify to

$$\begin{aligned} \pi_i^e &= 1 - (1 - p_z \rho^e)^{n_e - 1}, \\ \pi_i^a &= p_z + (1 - p_z) \left[1 - (1 - p_z \rho^a)^{n_e - 1}\right]. \end{aligned}$$

For instance, if researchers suspect that each ego–ego edge is possible with probability 1%, they can set $\rho^e = 0.01$.

Example 2 (Homogeneous number of edges)**.** A special case of Example 1 occurs by specifying the *expected number* of latent edges, separately for the ego–ego and alter–ego edges. Let $m^e \in \left\{1, \ldots, \binom{n_e}{2}\right\}$ be the expected number of ego–ego edges that are believed to be missing. Setting

$$\rho_{ij}^e = \rho^e(m^e) = \frac{m^e}{\binom{n_e}{2}},$$

weights equally each ego–ego edge probability and yield m^e expected ego–ego edges.
There are $n_a(n_e - 1)$ possible missing alter–ego edges. If m^a such edges are expected to be missing, set the probability of each edge to

$$\rho_{ij}^a = \rho^a(m^a) = \frac{m^a}{n_a(n_e - 1)}.$$

For instance, assume researchers suspect that each alter is connected to $k^a \geq 0$ additional egos on average. This can be represented by selecting $m^a = k^a \times n_a$. Similarly, if each ego is believed to be connected to $k^e \geq 0$ additional egos on average, then $m^e = \frac{k^e \times n_e}{2}$.

Example 3 (Heterogeneous probabilities)**.** Examples 1-2 can be extended to incorporate unit-level characteristics through covariates $\boldsymbol{X}_i$. A well-known result in network science is homophily (McPherson et al., 2001), which states that units with similar characteristics are more likely to interact. The covariates similarity can be captured through a distance function. Common examples include

- L^p norm: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = \|\boldsymbol{x}_i - \boldsymbol{x}_j\|_p$.
- Cosine distance: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 1 - \frac{\boldsymbol{x}_i^T \boldsymbol{x}_j}{\|\boldsymbol{x}_i\| \|\boldsymbol{x}_j\|}$.
- Jaccard distance: $d(\boldsymbol{x}_i, \boldsymbol{x}_j) = 1 - \frac{\|\boldsymbol{x}_i \cap \boldsymbol{x}_j\|}{\|\boldsymbol{x}_i \cup \boldsymbol{x}_j\|}$.

For $\gamma^e \geq 0$, define similarity weights using the distances $w_{ij}^e = \exp\left(-\gamma^e d(\boldsymbol{X}_i, \boldsymbol{X}_j)\right)$ or through the inner product $w_{ij}^e = \exp\left(\gamma^e \boldsymbol{X}_i^T \boldsymbol{X}_j\right)$. Then, set ego–ego edge probabilities to

$$\rho_{ij}^e = \frac{w_{ij}^e}{\sum_{j \in \mathcal{R}_e^i} w_{ij}^e}.$$

In both cases, γ^e is the sensitivity paramaters which controls the influence of covariate similarity. Taking $\gamma^e \to 0$ yields uniform probabilities $\rho_{ij}^e \to \frac{1}{n_e - 1}$, while larger γ^e concentrates mass on more similar nodes. Analogous definitions apply for alter–ego edges with weights w_{ij}^a, probabilities ρ_{ij}^a, and a sensitivity parameter $\gamma^a \geq 0$.

Example 4 (Heterogeneous number of missing edges)**.** Examples 2–3 can be combined to allow pair-specific heterogeneous probabilities while fixing the expected number of missing edges. Let w_{ij}^e and w_{ij}^a be the weights defined in Example 3 with $\gamma^e = \gamma^a = 1$. Define the total sum of weights among ego–ego and alter–ego pairs with missing edges,

$$W^e = \sum_{i < j \in \mathcal{R}_e} w_{ij}^e, \qquad W^a = \sum_{i \in \mathcal{R}_a} \sum_{j \in \mathcal{R}_e^{e(i)}} w_{ij}^a.$$

Similarly to Example 2, specify the expected numbers m^e and m^a of missing ego–ego and alter–ego edges, and set

$$\rho_{ij}^e = \frac{m^e}{W^e} w_{ij}^e, \qquad \rho_{ij}^a = \frac{m^a}{W^a} w_{ij}^a.$$

This ensure that the expected number of missing edges are m^e and m^a, while allowing for heterogeneous edge probabilities. Setting all weights (w_{ij}^e and w_{ij}^a) to one recovers Example 2.

3.4 Running the sensitivity analysis in practice

To assess the the impact of contamination on the causal estimates, researchers can utilize the bias-corrected estimators (5)-(6) for the indirect effects and (7)-(8) for the direct effects. The bias-corrected estimators depend on the sensitivity parameters that define the edge-level probabilities (9) and the parameter κ for the direct effects. Examples 1-4 provide practical guidelines on how to specify these parameters in practice. As the the true values of such parameters are unknown and can not be directly estimated from the data, researchers can resort to sensitivity analysis by exploring a range of plausible values. We consider here two approaches: grid sensitivity analysis (GSA) and probabilistic bias analysis (PBA).

In GSA, researchers specify a grid of values for the sensitivity parameters and compute the bias-corrected estimates for each combination of parameters (Greenland, 1996; Rosenbaum, 2005). That results in a grid of bias-corrected estimates. Such estimates can be coupled with

the associated confidence interval given each value of the sensitivity parameters in the specified grid.

In PBA, researchers specify a prior distribution for the sensitivity parameters and compute the distribution of bias-corrected estimates over the priors (Greenland, 2005; Lash et al., 2014; Fox, 2021). In practice, this distribution is approximated using Monte Carlo by sampling from the prior distribution, and summarized through averages and intervals of the resulting bias-corrected estimates. This distribution of bias-corrected estimates accounts for uncertainty in the sensitivity parameters. We can also compute a distribution of estimates that accounts for both design uncertainty (over the treatment assignment distribution) and bias uncertainty (from the prior of the sensitivity parameters).

We now describe the GSA and PBA approaches in more details.

Grid sensitivity analysis. Given a grid of sensitivity parameters, the full procedure can be performed with either analytic variance estimators or nonparametric ego-network bootstrap. In the analytic case, the steps are as follows:

(i) (Optional.) Estimate the outcome models $\mu_i^a(f)$ and $\mu_i^e(z)$. We reccomend using cross-fitting via a two-fold split. That is, estimate the outcome models on one fold and predict on the other fold, and vice versa. These outcome models are estimated only once and used for all sensitivity parameter values.

(ii) For each parameter in the grid of sensitivity parameters do:

  (a) Compute bias-corrected estimates of IE (5)-(6) and DE (7)-(8).

  (b) Estimate the variance of each bias-corrected estimator (Appendix A).

  (c) Compute confidence intervals for each estimator using normal approximation.

Alternatively, the analytic variance estimation and the normal approximation in step (ii)(b)-(c) can be replaced by a nonparametric ego-network bootstrap. That is, resample with replacement n_e ego-networks from the observed data, compute the bias-corrected estimates for each bootstrap sample, and estimate the variance or confidence intervals using the empirical distribution of the bootstrap estimates. This bootstrap procedure can also be modified to account for uncertainty in the outcome models estimation by re-estimating the outcome models for each bootstrap sample.

Probabilistic bias analysis. Instead of specifying a grid of sensitivity parameters, researchers can specify a prior distribution for each parameter. For example, if researchers believe that on average 1% of the possible ego–ego edges are missing, they can set $\rho^e \sim \text{Beta}(a, b)$ with mean $\frac{a}{a+b} = 0.01$ in Example 1. Let $p(\cdot)$ be the prior distribution for the sensitivity parameters. In practice, researchers can specify a separate prior for each parameter.

PBA enables us to obtain a distribution of bias-corrected estimates over the prior of the sensitivity parameter. This distribution can be summarized, e.g., through average and percentiles. This distribution accounts only for the bias uncertainty (from prior of sensitivity parameters). Incorporating design uncertainty (from the distribution of $\boldsymbol{Z}$) can be achieved through random error simulation or bootstrapping (Greenland, 2005; Fox, 2021). We now describe the general structure of the PBA procedure. Let $B > 0$ be the number of Monte Carlo samples.

(i) For each $b = 1, \ldots, B$ do:

  (a) Sample sensitivity parameters from the prior p.

  (b) Compute the bias-corrected estimates $\widehat{IE}_{adj}^{(b)}$ and $\widehat{DE}_{adj}^{(b)}$, and estimate their variance using the analytic formulas (Appendix A).

  (c) Draw from the (approximate) random error distribution

$$\widehat{IE}_{adj}^{(b,rand)} \sim N\left(\widehat{IE}_{adj}^{(b)}, \widehat{\text{Var}}\left(\widehat{IE}_{adj}^{(b)}\right)\right),$$

  and similarly for the other estimators.

(ii) Summarize the distribution of bias-corrected estimates. For example, by computing the average $\frac{1}{B}\sum_{b=1}^{B}\widehat{IE}_{adj}^{(b,rand)}$ and empirical percentiles for intervals.

Including the outcome models can be achieved similarly to the GSA procedure by estimating them once using cross-fitting before the PBA steps. In addition, the analytic variance estimation and random error simulation in step (i)(b)-(c) can be replaced by a two-dimensional nonparametric bootstrap (Fox, 2021). That is, for each Monte Carlo sample $b = 1, \ldots, B$, resample with replacement n_e ego-networks from the observed data, and compute the bias-corrected estimates on the bootstrap sample. The resulting empirical distribution of the bootstrap estimates will account for both design and bias uncertainty, and can be summarized accordingly. We provide more technical details on the motivation behind the PBA procedure in Appendix C.

4 Simulation study

We conduct a simulation study to evaluate the performance of the bias-corrected estimators (5)-(6) and (7)-(8). Specifically, we assess the bias, variance estimation, and empirical coverage of the confidence intervals given that the true sensitivity parameters are known. The simulation study is based on a data generating process that mimics the settings of the HPTN-037 study (Latkin et al., 2009), which we analyze in Section 5. We generated $n_e = 200$ disjoint ego-networks with two alters each, which was the average number of alters per ego-network in HPTN-037, resulting in $n_a = 400$. Contamination betweeh the ego-networks was generated by specifiying the expected number of missing alter–ego and ego–ego edges, as in Examples 2 and 4, and drawing the latent edges independently. Specifically, we set the expected number of missing alter–ego edges to $m^a \in \{100, 200, 300\}$, and ego–ego edges to $m^e \in \{50, 150, 250\}$. The binary potential outcomes were generated according to the models

$$\begin{aligned}
\text{logit}\left(\Pr\left(Y_i(z,f) = 1 \mid i \in \mathcal{R}_e\right)\right) &= -2 + 2z + 0.5f + 2zf + \boldsymbol{X}_i^T\beta_e, \\
\text{logit}\left(\Pr\left(Y_i(0,f) = 1 \mid i \in \mathcal{R}_a\right)\right) &= -2 + 2f + \boldsymbol{X}_i^T\beta_a,
\end{aligned}$$

implying interaction sensitivity parameter $\kappa = 2$. The data-generating proccess of the covariates $\boldsymbol{X}_i$ and the specification of β_e and β_a are provided in Appendix D. Treatments were assigned according to a Bernoulli design with $p_z = 0.5$. In each of the 5×10^3 simulation replications, we run the ENRT trial by generating treatment assignment. We report here the results for indirect effects where contamination is generated using heterogeneous edge probabilities as in Example 4 with Euclidean distance and $\gamma^e = \gamma^a = 1$. Table 1 summarize the results, using both analytic and bootstrap variance estimation. We use the augmented estimator (6) with outcome models estimated by logistic regression via cross-fitting. When specification used is heterogeneous, the edge probabilities are correctly-specified, while using homogeneous specification uses the same probability for all edges as in Example 2, and is therefore misspecified. The naive specification corresponds to the uncorrected estimator (3). As the level of contamination (m_a) increase, the bias of the naive estimators increases and the coverage decreases substantially. In contrast, both bias-corrected estimators exhibit negligible bias and valid coverage across all scenarios, implying that even a misspecified model for the edge probabilities (homogeneous specification) can effectively correct for the bias. Moreover, across all scenarios, both analytic and bootstrap variance estimators over estimate the variance slightly. Results for direct effects, additional results, including homogeneous contamination settings and non-augmented estimators, are provided in Appendix D and exhibit similar patterns.

Table 1: Simulation results for indirect effects under heterogeneous contamination with augmented estimators. Error is the empirical bias, Coverage is the empirical coverage of the 95% confidence intervals, and SD/SE is the ratio of standard deviation of the estimates to the mean standard errors. Specification "Heterogeneous" uses the correct edge probabilities, "Homogeneous" uses the same probability for all edges, and "Naive" is the uncorrected estimator.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_a = 100, IE = 0.358$	Heterogeneous	Analytic	0.016	0.978	0.774
		Bootstrap	0.016	0.978	0.770
	Homogeneous	Analytic	0.015	0.979	0.771
		Bootstrap	0.015	0.981	0.770
	Naive	Analytic	-0.029	0.951	0.772
		Bootstrap	-0.029	0.951	0.773
$m_a = 200, IE = 0.335$	Heterogeneous	Analytic	0.006	0.977	0.839
		Bootstrap	0.006	0.977	0.841
	Homogeneous	Analytic	0.008	0.977	0.828
		Bootstrap	0.008	0.978	0.829
	Naive	Analytic	-0.068	0.678	0.829
		Bootstrap	-0.068	0.678	0.829
$m_a = 300, IE = 0.318$	Heterogeneous	Analytic	-0.004	0.972	0.906
		Bootstrap	-0.004	0.972	0.904
	Homogeneous	Analytic	0.003	0.971	0.902
		Bootstrap	0.003	0.970	0.904
	Naive	Analytic	-0.097	0.453	0.904
		Bootstrap	-0.098	0.451	0.902

5 Data analysis

HPTN 037 was a ENRT that evaluated the efficacy of a peer education intervention in reducing HIV risk behaviors among people who inject drugs and their drug and sexual networks (Latkin et al., 2009). Egos were randomized ($p_z = 0.5$) to receive a peer education training and were encouraged to share the HIV risk reduction information with their drug and sexual network members (alters). Egos in the control arm received standard HIV prevention counseling. The study was conducted between 2002 and 2006 in Philadelphia, Pennsylvania and Chiang Mai, Thailand. The study recruited 414 egos (index participants) who were asked to recruit at least one drug or sex network member (alter) into the study. Researchers enforced that alters were not also egos, and that each alter was connected to only one ego (Latkin et al., 2009). Following previous analyses (Buchanan et al., 2018, 2024; Chao et al., 2025), we limit our analysis to the Pennsylvania site and to a 12-month follow-up, yielding a sample of $n_e = 150$ egos and $n_a = 263$ alters, with an average of 2 alters per ego-network. The primary outcome is a binary indicator of any injection-related risk behavior in the past month, including sharing injection equipment, front and back loading, and injecting with people not well known or in a public space.

Contamination across ego-networks ia concern, as egos and alters may have social connections outside of those sampled in the study. That is, alters might inject drugs or have sexual relations with egos from other ego-networks, or egos might be connected to each other. Using recall data collected after the trial, Simmons et al. (2015) found evidence of contamination. Chao et al. (2025) developed a bias-correction method to adjust for (homogeneous) exposure misclassification, only for the indirect effects, building on the recall data and the unverifiable assumption of non-overlapping ego-networks. That is, Chao et al. (2025) assumed that, in the population network, alters can only be connected to one ego, and egos can not be connected to each other.

We relax this assumption and apply our proposed sensitivity analysis methods to assess the impact of contamination on both indirect and direct effects estimates. For both alter–ego and ego–ego edges, we specify the expected number of missing edges as in Examples 2 and 4. Covariates used in the analysis include risk behaviors at baseline, age, gender, and race. For the heterogeneous edge probabilities specification (Example 4), we use the Euclidean distance

and set $\gamma^e = \gamma^a = 1$. For the direct effects, we specify values larger than 1 to the interaction sensitivity parameter κ, as we expect that the direct effect is larger among egos exposed to other treated egos. We apply both the grid sensitivity analysis (GSA) and probabilistic bias analysis (PBA) described in Section 3.4.

5.1 Grid sensitivity analysis

For the indirect effects, we consider a grid of expected number of missing edges with $m^a \in [10, 400]$, representing the suspiction that, in the homogeneous case, each alter is connected to up to 1.5 additional egos on average. For the ego–ego edges, we consider $m^e \in [10, 300]$ and $\kappa \in [1.1, 3]$ reflecting the suspicion that direct effects among egos connected to other treated egos is up to three times larger than the effects among egos that are not connected to other treated egos. We report here the results for the estimators (5) and (7) with both homogeneous and heterogeneous edge probabilities specifications and nonparametric ego-network bootstrap with 10^4 resamples. Results for the augmented estimators (6) and (8) are provided in Appendix E, and were consistent with the not-augmented results shown here.

Figures 2 and 3 summarize the GSA results for indirect and direct effects, respectively. For indirect effects, both homogeneous and heterogeneous specifications yield similar results. The results suggest that, for instance, if each alter is connected to up to 1 additional ego on average ($m^a = 263$) in the homogeneous case, the bias-corrected estimate is around -0.225 compared to the naive estimate of -0.137, indicating a substantial bias due to contamination. For the direct effects, the bias-corrected estimates are closer to zero than the naive estimate of -0.147 across the entire grid of sensitivity parameters for both homogeneous and heterogeneous specifications. For example, if each ego is connected to up to 1 additional ego on average ($m^e = 75$) and $\kappa = 2$, the bias-corrected estimate is around -0.1 in the homogeneous case. That suggests that assuming no-contamination leads to over-estimation of the direct effect. However, both indirect and direct bias-corrected estimates remain statistically insignificant at the 5% level across the entire grid of sensitivity parameters for both homogeneous and heterogeneous specifications.

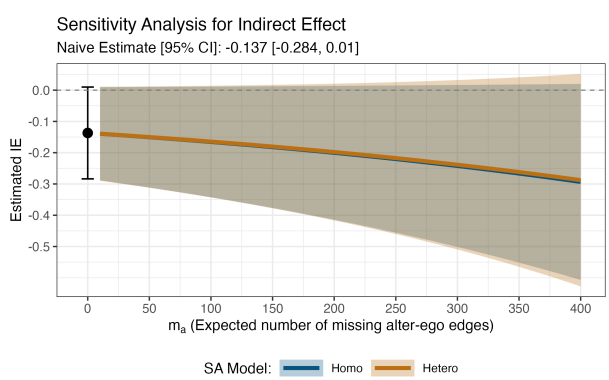

Figure 2: Grid sensitivity analysis results for indirect effects using the estimator (5). The x-axis represents the expected number of missing alter–ego edges m^a, while the y-axis represents the estimates values. Results are shown for both homogeneous (blue) and heterogeneous (orange) edge probabilities specifications.

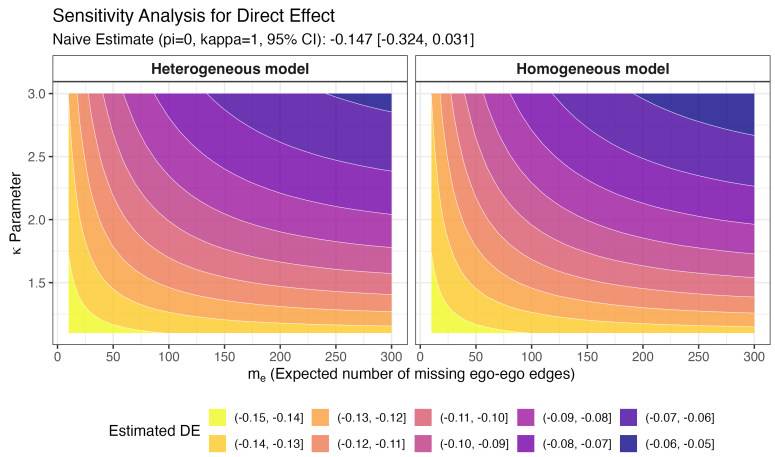

Figure 3: Grid sensitivity analysis results for direct effects using the estimator (7). The x-axis represents the expected number of missing ego–ego edges m^e, while the y-axis represents κ, the interaction sensitivity parameter. Contour lines and color shading represent the estimates values. Results are shown for both heterogeneous (left) and homogeneous (right) edge probabilities specifications.

5.2 Probabilistic bias analysis

For the PBA, we specify prior distributions for the sensitivity parameters, representing different degrees of certainty about their values. For m^a and m^e we consider three prior distributions. The first, is discrete uniform distribution over the ranges used in the GSA: $m^a \sim \text{Uniform}\{10, 400\}$ and $m^e \sim \text{Uniform}\{10, 300\}$, representing maximum entropy, but capped right tail. The second is a Poisson distribution with mean equal to half of the maximum value in the GSA grid: $m^a \sim \text{Poisson}(200)$ and $m^e \sim \text{Poisson}(150)$, representing moderate uncertainty with unbounded but thin tail. The third is a Negative Binomial (NB) distribution with mean equal to half of the maximum value in the GSA grid and size parameter equal to 10: $m^a \sim \text{NB}(10, 1/21)$ and $m^e \sim \text{NB}(10, 1/16)$, representing heavier tail than the Poisson. For the interaction sensitivity parameter κ, we consider a continuous uniform prior $\kappa \sim \text{Uniform}[1, 3]$, and a log-normal distribution with mean equal to $\log(2)$ and standard deviation equal to 0.2 on the log scale, resulting in a heavier right tail than the uniform. We run the PBA procedure with $B = 10^4$ Monte Carlo samples with the non-parameteric ego-network bootstrap estimator.

Figure 4 summarize the PBA results for indirect and direct effects with the not-augmented estimators. For indirect effects, the three prior distributions for m^a yield similar point estimates, with slighly larger magnitude for the heterogeneous specification. The uniform prior results in the widest intervals, while the Poisson prior had the narrowest intervals. All bias-corrected estimates are lower (point estimates around -0.2) than the naive estimate of -0.137, suggesting that contamination leads to under-estimation of the indirect effect. For the direct effects, all prior distributions for m^e and κ yield similar point estimates. The uniform priors yielded the widest intervals. All bias-corrected estimates are closer to zero (point estimates around -0.1) than the naive estimate of -0.147, suggesting that contamination leads to over-estimation of the direct effect. Nevertheless, across all priors and edge probabilities specifications, the indirect and direct 95% PBA intervals include zero.

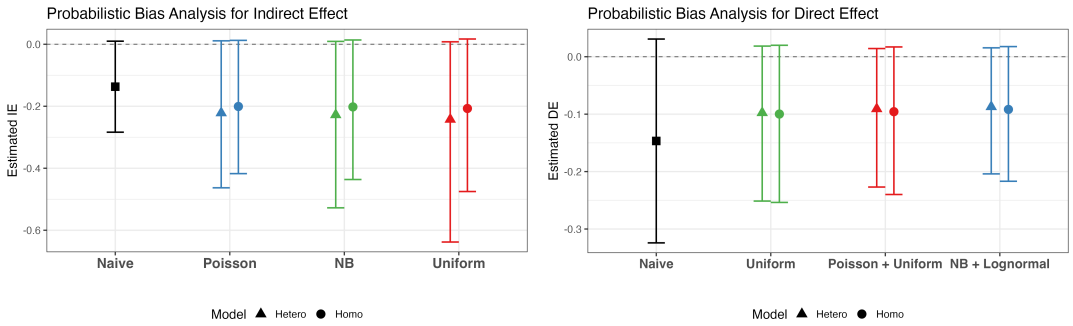

Figure 4: Probabilistic bias analysis results for indirect effects (left) using the estimator (5), and direct effects (right) using the estimator (7). Results are shown for both homogeneous and heterogeneous edge probabilities specifications. Colors represent different prior distributions for the expected number of missing edges and the interaction sensitivity parameter κ.

6 Discussion

In this paper, we developed sensitivity analysis methods to assess the impact of contamination between ego-networks on the causal estimates in ego-centric network randomized trials. Such trials are increasingly used to evaluate interventions in hard-to-reach populations, where full network information is often unavailable or difficult to obtain. Our methods build on a general formulation of contamination as exposure misclassification due to missing network edges, and provide bias-corrected estimators for both indirect and direct effects. The bias-corrected estimators depend on sensitivity parameters that characterize the contamination process at the edge-level and the interaction magnitude between treatments and exposures for the direct effects. We provided practical guidelines for specifying these sensitivity parameters and described two sensitivity analysis approaches: grid sensitivity analysis and probabilistic bias analysis. We illustrated our methods through a simulation study and an analysis of the HPTN 037 trial, showing that the indirect effects may be under-estimated and the direct effects over-estimated when contamination is ignored.

We assumed a two-level exposure mappings specification, indicating whether a unit is connected to at least one treated neighbor (Assumption 2). This specification makes senses when there is no contamination between ego-networks or when the treatment effect is saturated at one treated neighbor. That is, when exposure to additional treated neighbors does not further affect the outcome. In Appendix B, we extend our methods to a three-level exposure mapping specification, distinguishing between no treated neighbors, one treated neighbor, and two or more treated neighbors. In this case, for example, alter with observed exposure level of one treated neighbor may have true exposure level of one or two-and-more treated neighbors. For the same indirect effects estimand (1), we derive the bias expressions for the naive estimator (3). We further provide bias-corrected estimators that depend on additional sensitivity parameter that capture the marginal effect of exposure, analogously to κ in Assumption 4. We show that the the two-level exposure mapping model is a special case of the three-level model with saturation of effects after the first exposure.

Ego-centric network randomized trial is a promising design that allows for the estimation of spillovers and indirect effects without the requirement for sociocentric network measurements. By that, it enables researchers to estimate causal effects under interference with limited network data. However, extending these designs to observational studies where units are sampled from a larger network (e.g., through ego-centric sampling), but treatments are not randomly assigned by the researchers is problematic. The primary concern is that non-recruited units can be treated as well, leading to treatment interference between recruited and non-recruited units. Since information on the edges connecting such units in the population is rarely available, addressing interference in observational studies with ego-centric sampling remains a significant challenge.

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A Proofs

A.1 Indirect effects

Recall that the HT estimator of the indirect effects based on the observed data (3) is

$$\widehat{IE} = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right]$$

By the assumed experimental design (Assumption 1), the probability that an alter is assigned to treatment is zero, that is, $\Pr(Z_i = 0 \mid i \in \mathcal{R}_a) = 0$. Also, Z_i and $\widetilde{F}_i = F(\boldsymbol{Z}_{-i}, \widetilde{\boldsymbol{A}})$ are independent. Thus, the event $\mathbb{I}\{\widetilde{F}_i = f\}$ is equal with probability one to $\mathbb{I}\{Z_i = 0, \widetilde{F}_i = f\}$ among the alters $i \in \mathcal{R}_a$. Starting with the first inner term, for $i \in \mathcal{R}_a$,

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z} \right] &\overset{(i)}{=} \frac{1}{p_z} \mathbb{E}_{\boldsymbol{Z}} \left[\mathbb{I}\{F_i = 1, \widetilde{F}_i = 1\} Y_i(0,1) + \mathbb{I}\{F_i = 0, \widetilde{F}_i = 1\} Y_i(0,0) \right] \\
&\overset{(ii)}{=} \frac{\Pr(F_i = 1, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) Y_i(0,1)}{p_z} \\
&\quad + \frac{\Pr(F_i = 0, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a) Y_i(0,0)}{p_z} \\
&\overset{(iii)}{=} \Pr(F_i = 1 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) Y_i(0,1) \\
&\quad + \Pr(F_i = 0 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) Y_i(0,0) \\
&\overset{(iv)}{=} Y_i(0,1),
\end{aligned} \tag{A.1}$$

where (i) follows since $\mathbb{I}\{\widetilde{F}_i = f\}$ is equal with probability one to $\mathbb{I}\{Z_i = 0, \widetilde{F}_i = f\}$ and consistency (Assumption 3) connect the observed outcomes Y_i to the potential outcomes $Y_i(z, f)$; (ii) from taking expectation with respect to the design $\boldsymbol{Z}$; (iii) since $p_z = \Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)$ and $\frac{\Pr(F_i = f, \widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 1 \mid i \in \mathcal{R}_a)} = \Pr(F_i = f \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a)$; and (iv) since $\Pr(F_i = 1 \mid \widetilde{F}_i = 1, i \in \mathcal{R}_a) = 1$. To see this, note that $\sum_{j \neq i} Z_j \widetilde{A}_{ij} \leq \sum_{j \neq i} Z_j A_{ij}$ with probability one, which implies that $\widetilde{F}_i \leq F_i$ with probability one as well. Since F_i is binary, if $\widetilde{F}_i = 1$, it must also be $F_i = 1$.

Moving to the second inner term, a similar derivation yields

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right] &= \Pr(F_i = 1 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) Y_i(0,1) \\
&\quad + \Pr(F_i = 0 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) Y_i(0,0).
\end{aligned}$$

By Bayes' rule,

$$\begin{aligned}
\Pr(F_i = 0 \mid \widetilde{F}_i = 0, i \in \mathcal{R}_a) &= \frac{\Pr(\widetilde{F}_i = 0 \mid F_i = 0, i \in \mathcal{R}_a) \Pr(F_i = 0 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 0 \mid i \in \mathcal{R}_a)} \\
&= \frac{\Pr(F_i = 0 \mid i \in \mathcal{R}_a)}{\Pr(\widetilde{F}_i = 0 \mid i \in \mathcal{R}_a)} \\
&= \frac{1 - \pi_i^a}{1 - p_z},
\end{aligned}$$

as $\Pr(\widetilde{F}_i = 0 \mid F_i = 0, i \in \mathcal{R}_a) = 1$ since $\widetilde{F}_i \leq F_i$ with probability one. This yields

$$\mathbb{E}_{\boldsymbol{Z}} \left[\frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z} \right] = \frac{(\pi_i^a - p_z) Y_i(0,1) + (1 - \pi_i^a) Y_i(0,0)}{1 - p_z}. \tag{A.2}$$

Consequently, combining (A.1) and (A.2),

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\left[Y_i(0,1) - \frac{(\pi_i^a - p_z)\,Y_i(0,1) + (1-\pi_i^a)\,Y_i(0,0)}{1-p_z}\right] \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-\pi_i^a}{1-p_z}\left[Y_i(0,1) - Y_i(0,0)\right] \\
&= IE + \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{p_z-\pi_i^a}{1-p_z}\left[Y_i(0,1) - Y_i(0,0)\right],
\end{aligned} \tag{A.3}$$

which completes the proof of the first part of Proposition 1.

The bias-corrected estimator is

$$\widehat{IE}_{adj} = \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\left[\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}\right].$$

As the weights $\frac{1-p_z}{1-\pi_i^a}$ are fixed and therefore independent of the design $\boldsymbol{Z}$, taking the expectation $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right]$ will result in the second row of (A.3) where the terms $\frac{1-\pi_i^a}{1-p_z}$ and $\frac{1-p_z}{1-\pi_i^a}$ are multiplied and therefore canceled. That implies $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right] = IE$ and completes the proof of Proposition 2.

Variance estimation. We derive variance estimators based on the augmented estimator (6). Write

$$\begin{aligned}
\widehat{IE}_{adj}^{aug} &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\left[\frac{\mathbb{I}\{\widetilde{F}_i=1\}\left(Y_i-\mu_i^a(1)\right)}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i=0\}\left(Y_i-\mu_i^a(0)\right)}{1-p_z} + \{\mu_i^a(1)-\mu_i^a(0)\}\right] \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\phi_i^a \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^a\phi_i^a,
\end{aligned}$$

where $\alpha_i^a = \frac{1-p_z}{1-\pi_i^a}$. We assume that the outcome models $\mu_i^a(f)$ are estimated through cross-fitting (Chernozhukov et al., 2018) and therefore assumed to be independent of $\boldsymbol{Z}$. We ignore the sampling uncertainty associated with estimating the outcome models. Accounting for such uncertainty is possible through the nonparametric ego-network bootstrap procedure.

Alters in the same ego-network have the same observed exposure. Recall that $e(i) \in \mathcal{R}_e$ is the ego of alter $j \in \mathcal{R}_a$. We can write the adjusted IE estimator (6) as

$$\begin{aligned}
\widehat{IE}_{adj}^{aug} &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^a\phi_i^a \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_e}\sum_{j\in\mathcal{R}_a:e(j)=i}\alpha_j^a\phi_j^a,
\end{aligned}$$

which is the sum over all ego-networks of the sum of alters estimates within each ego-network. As estimates of alters in the same ego-network are likely correlated (often represented as the intra-cluster correlation coefficient in cluster randomized trials), we estimate the variance of $\widehat{IE}_{adj}^{aug}$ using the cluster-robust empirical variance estimator. For ego $i \in \mathcal{R}_e$, write

$$T_i = \sum_{j\in\mathcal{R}_a:e(j)=i}\alpha_j^a\phi_j^a,$$

such that

$$\widehat{IE}_{adj}^{aug} = \frac{1}{n_a}\sum_{i\in\mathcal{R}_e}T_i.$$

Let $\overline{T} = \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} T_i$. Thus, $\widehat{IE}^{aug}_{adj} = \frac{n_e}{n_a}\overline{T}$. The cluster-robust empirical variance estimator is

$$\widehat{Var}\left(\widehat{IE}^{aug}_{adj}\right) = \frac{n_e}{n_a^2(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(T_i - \overline{T}\right)^2.$$

Alternatively, it is also possible to estimate the variance of $\widehat{IE}^{aug}_{adj}$ by nonparametric network bootstrap, where we resample ego-networks.

Attenuation towards the null. Eq. (11) implies that $\pi_i^a \geq p_z$, therefore $\frac{1-p_z}{1-\pi_i^a} \geq 1$. If $\pi_i^a = \pi^a$ for all $i \in \mathcal{R}_a$, that is, the same for all alters, then $\widehat{IE}_{adj} = \frac{1-p_z}{1-\pi^a}\widehat{IE}$. Thus, $\text{sign}(\widehat{IE}_{adj}) = \text{sign}(\widehat{IE})$ and

$$\begin{cases}\widehat{IE}_{adj} \geq \widehat{IE}, & \widehat{IE} > 0,\\ \widehat{IE}_{adj} \leq \widehat{IE}, & \widehat{IE} < 0.\end{cases}$$

Moreover, if $Y_i(0,1) \geq Y_i(0,0)$ for all $i \in \mathcal{R}_a$ such that $IE \geq 0$, then since $p_z - \pi_i^a \leq 0$, from (A.3), we have $0 \leq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \leq IE$. Vice versa, if $Y_i(0,1) \leq Y_i(0,0)$ for all $i \in \mathcal{R}_a$, then $IE \leq 0$ and $0 \geq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \geq IE$. That is, the estimator $\widehat{IE}$ is, in expectation, attenuated towards the null of no indirect effects.

Exposure probabilities of alters. For alters $i \in \mathcal{R}_a$, $F_i = 1$ if its ego $e(i)$ or at least one of the other recruited egos connected to i in the population network $\boldsymbol{A}$ is treated. We have

$$\begin{aligned}\pi_i^a &= \Pr(F_i = 1 \mid i \in \mathcal{R}_a)\\ &= \Pr(F_i = 1 \mid Z_{e(i)} = 1, i \in \mathcal{R}_a)\Pr(Z_{e(i)} = 1 \mid i \in \mathcal{R}_a)\\ &\quad + \Pr(F_i = 1 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a)\Pr(Z_{e(i)} = 0 \mid i \in \mathcal{R}_a),\end{aligned}$$

but $\Pr(F_i = 1 \mid Z_{e(i)} = 1, i \in \mathcal{R}_a) = 1$ and $\Pr(Z_{e(i)} = 1 \mid i \in \mathcal{R}_a) = p_z\mathbb{I}\{e(i) \in \mathcal{R}_e\}$ by Assumptions 1 and 2. Since only recruited egos have a non-zero probability of being assigned to treatment, then for alter $i \in \mathcal{R}_a$, $F_i = \sum_{j\neq i} Z_j A_{ij} = \sum_{j\in\mathcal{R}_e} Z_j A_{ij}$ with probability one. Thus, $F_i = 0 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a$ equal with probability one to $\mathbb{I}\{\sum_{j\in\mathcal{R}_e\setminus e(i)} Z_j A_{ij} = 0\}$. Thus,

$$\Pr(F_i = 0 \mid Z_{e(i)} = 0, i \in \mathcal{R}_a) = \prod_{j\in\mathcal{R}_e\setminus\{e(i)\}}\left(1 - p_z\rho_{ij}^a\right),$$

which yield (11).

Extension to relative risk estimand and estimators. In the case of binary outcomes, we can also consider the relative risk (RR) estimand

$$IE^{RR} = \frac{\sum_{i\in\mathcal{R}_a} Y_i(0,1)}{\sum_{i\in\mathcal{R}_a} Y_i(0,0)}. \tag{A.4}$$

Analogously to (3), the estimator based on the observed data is therefore

$$\widehat{IE}^{RR} = \frac{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}}. \tag{A.5}$$

We saw in (A.1) that the numerator in (A.5) is an unbiased estimator of the numerator in (A.4). However, from (A.2) the denominator is biased. Rearranging (A.2),

$$\frac{1-p_z}{1-\pi_i^a}\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{\widetilde{F}_i = 0\}Y_i}{1-p_z}\right] = Y_i(0,0) + \frac{\pi_i^a - p_z}{1-\pi_i^a}Y_i(0,1).$$

Thus, using (A.1), we can unbiasedly estimate $Y_i(0,0)$ through

$$\frac{1-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}-\frac{\pi_i^a-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}.$$

That motivates the adjusted estimator

$$\widehat{IE}_{adj}^{RR}=\frac{\sum_{i\in\mathcal{R}_a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}{\sum_{i\in\mathcal{R}_a}\frac{1-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=0\}Y_i}{1-p_z}-\frac{\pi_i^a-p_z}{1-\pi_i^a}\frac{\mathbb{I}\{\widetilde{F}_i=1\}Y_i}{p_z}}. \tag{A.6}$$

Being a ratio estimator, (A.6) is biased, but the bias can be easily bounded (Särndal et al., 2003).

A.2 Direct effects

Recall that the HT estimator of direct effects based on the observed data (4) is

$$\widehat{DE}=\frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}-\frac{\mathbb{I}\{Z_i=0\}Y_i}{1-p_z}\right].$$

Also, the observed exposure of all egos is zero $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)=1$. By the assumed expermential design (Assumption 1), the event $\mathbb{I}\{Z_i=z\}$ is equal with probability one to $\mathbb{I}\{Z_i=z,\widetilde{F}_i=0\}$ among the egos $i\in\mathcal{R}_e$. The expectation of the first inner term of $\widehat{DE}$, for $i\in\mathcal{R}_e$, is as follows.

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}\right] &\overset{(i)}{=} \mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1,\widetilde{F}_i=0\}Y_i}{p_z}\right]\\
&\overset{(ii)}{=} \frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{Z_i=1,F_i=1,\widetilde{F}_i=0\}Y_i(1,1)\right]\\
&\quad+\frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{Z_i=1,F_i=0,\widetilde{F}_i=0\}Y_i(0,0)\right]\\
&\overset{(iii)}{=} \Pr(F_i=1,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)Y_i(1,1)\\
&\quad+\Pr(F_i=0,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)Y_i(1,0)\\
&\overset{(iv)}{=} \frac{\Pr(F_i=1,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}Y_i(1,1)\\
&\quad+\frac{\Pr(F_i=0,\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}Y_i(1,0)\\
&=\Pr(F_i=0\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)Y_i(1,1)\\
&\quad+\Pr(F_i=1\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)Y_i(1,0),
\end{aligned}$$

where (i) follows from the explanation described above; (ii) from consistency (Assumption 3); (iii) since Z_i is independent from both F_i and $\widetilde{F}_i$ by Assumption 1, and $\Pr(Z_i=1\mid i\in\mathcal{R}_e)=p_z$; and (iv) since $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)=1$. By Bayes' rule,

$$\begin{aligned}
\Pr(F_i=1\mid\widetilde{F}_i=0,i\in\mathcal{R}_e)&=\frac{\Pr(\widetilde{F}_i=0\mid F_i=1,i\in\mathcal{R}_e)\Pr(F_i=1\mid i\in\mathcal{R}_e)}{\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)}\\
&=\Pr(F_i=1\mid i\in\mathcal{R}_e),
\end{aligned}$$

since $\widetilde{F}_i=0$ with probability one for $i\in\mathcal{R}_e$, which implies that $\Pr(\widetilde{F}_i=0\mid i\in\mathcal{R}_e)$ and the conditional probability $\Pr(\widetilde{F}_i=0\mid F_i=f,i\in\mathcal{R}_e)$ equal to one. Let $\pi_i^e=\Pr(F_i=1\mid i\in\mathcal{R}_e)$. Plugging this into the above expectation yields

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z}\right]=\pi_i^eY_i(1,1)+(1-\pi_i^e)Y_i(1,0). \tag{A.7}$$

Similarly, for $z = 0$ we have

$$\mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{Z_i = 0\}Y_i}{1 - p_z}\right] = \pi_i^e Y_i(0,1) + (1 - \pi_i^e)Y_i(0,0). \tag{A.8}$$

Plugging (A.7) and (A.8) into $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right]$ yields

$$\begin{aligned}\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\left[\pi_i^e\left\{Y_i(1,1) - Y_i(0,1)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\pi_i^e\left[\left\{Y_i(1,1) - Y_i(0,1)\right\} - \left\{Y_i(1,0) - Y_i(0,0)\right\}\right],\end{aligned} \tag{A.9}$$

which completes the second part of Proposition 1.

Now, under Assumption 4, $Y_i(1,1) - Y_i(0,1) = \kappa\left\{Y_i(1,0) - Y_i(0,0)\right\}$. Recall that the adjusted DE estimator (7) is

$$\widehat{DE}_{adj} = \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\frac{\mathbb{I}\{Z_i=1\}Y_i}{p_z} - \frac{\mathbb{I}\{Z_i=0\}Y_i}{1-p_z}\right].$$

Since both κ and π_i^e are fixed and therefore independent of $\boldsymbol{Z}$, we can use (A.7), (A.8), and (A.9) to obtain

$$\begin{aligned}\mathbb{E}\left[\widehat{DE}_{adj}\right] &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\pi_i^e\left\{Y_i(1,1) - Y_i(0,1)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &\overset{(i)}{=} \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\pi_i^e\kappa\left\{Y_i(1,0) - Y_i(0,0)\right\} + (1-\pi_i^e)\left\{Y_i(1,0) - Y_i(0,0)\right\}\right] \\ &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}Y_i(1,0) - Y_i(0,0) \\ &= DE,\end{aligned}$$

where in (i) we used Assumption 4.

Variance estimation. We derive variance estimators based on the augmented estimator (8). Write

$$\begin{aligned}\widehat{DE}_{adj}^{aug} &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e}\frac{1}{1+\pi_i^e(\kappa-1)}\left[\frac{\mathbb{I}\{Z_i=1\}\left(Y_i - \mu_i^e(1)\right)}{p_z} - \frac{\mathbb{I}\{Z_i=0\}\left(Y_i - \mu_i^e(0)\right)}{1-p_z} + \left\{\mu_i^e(1) - \mu_i^e(0)\right\}\right], \\ &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1}{1+\pi_i^e(\kappa-1)}\phi_i^e \\ &= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^e\phi_i^e.\end{aligned}$$

We assume that the outcome models $\mu_i^e(z)$ are estimated through cross-fitting (Chernozhukov et al., 2018) and therefore assumed to be independent of $\boldsymbol{Z}$. We ignore the sampling uncertainty associated with estimating the outcome models. We estimate the variance using the empirical estimate

$$\begin{aligned}\widehat{Var}\left(\widehat{DE}_{adj}^{aug}\right) &= \frac{1}{n_e(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(\alpha_i^e\phi_i^e - \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\alpha_i^e\phi_i^e\right)^2 \\ &= \frac{1}{n_e(n_e-1)}\sum_{i\in\mathcal{R}_e}\left(\alpha_i^e\phi_i^e - \widehat{DE}_{adj}^{aug}\right)^2.\end{aligned}$$

Alternatively, it is also possible to estimate the variance of $\widehat{DE}_{adj}^{aug}$ by nonparametric network bootstrap, where we resample ego-networks.

Attenuation and interplay between κ and π_i^e. Consider first the case homogeneous exposure probabilities $\pi_i^e = \pi^e$ for all $i \in \mathcal{R}_e$. Then, $\widehat{DE}_{adj} = \frac{1}{2+\pi^e(\kappa-1)}\widehat{DE}$. Thus, if $1+\pi^e(\kappa-1) > 1 \Rightarrow \kappa > 1 - \frac{1}{\pi^e}$, then $\text{sign}(\widehat{DE}_{adj}) = \text{sign}(\widehat{DE})$. Otherwise, if $\kappa < 1 - \frac{1}{\pi^e}$, then $\text{sign}(\widehat{DE}_{adj}) = -\text{sign}(\widehat{DE})$. However, for example, if $\pi^e \approx 0.01$, $1 - \frac{1}{\pi^e} \approx -99$, and it is very unlikely that κ is smaller than that value in practice. So in general, we expect that $\text{sign}(\widehat{DE}_{adj}) = \text{sign}(\widehat{DE})$.

Moreover, if $\kappa > 1$ then

$$\begin{cases} \widehat{DE}_{adj} < \widehat{DE}, & \widehat{DE} > 0, \\ \widehat{DE}_{adj} > \widehat{DE}, & \widehat{DE} < 0, \end{cases}$$

so the adjusted estimator is, in magnitude, closer to zero than the unadjusted estimator. Vice versa, if $\kappa \in \left(1 - \frac{1}{\pi^e}, 1\right)$ then

$$\begin{cases} \widehat{DE}_{adj} > \widehat{DE}, & \widehat{DE} > 0, \\ \widehat{DE}_{adj} < \widehat{DE}, & \widehat{DE} < 0, \end{cases}$$

so the adjusted estimator is, in magnitude, further away from zero than the unadjusted estimator.

Under Assumption 4, we have $Y_i(1,1) - Y_i(0,1) = \kappa\{Y_i(1,0) - Y_i(0,0)\}$. Thus, we can write (A.9) as

$$\begin{aligned} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \pi_i^e \left[\{Y_i(1,1) - Y_i(0,1)\} - \{Y_i(1,0) - Y_i(0,0)\}\right] \\ &= DE + \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \pi_i^e(\kappa-1)\left[Y_i(1,0) - Y_i(0,0)\right] \\ &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_e} \left[1 + \pi_i^e(\kappa-1)\right]\left[Y_i(1,0) - Y_i(0,0)\right] \end{aligned}$$

Since $\pi_i^e \geq 0$, we have

- If $Y_i(1,0) \geq Y_i(0,0)$ for all $i \in \mathcal{R}_e$, then $DE \geq 0$ and

$$\begin{cases} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq DE \geq 0, & \kappa > 1, \\ 0 \leq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq DE, & \kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right), \\ \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq 0, & \kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}. \end{cases}$$

- If $Y_i(1,0) \leq Y_i(0,0)$ for all $i \in \mathcal{R}_e$, then $DE \leq 0$ and

$$\begin{cases} \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \leq DE \leq 0, & \kappa > 1, \\ 0 \geq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq DE, & \kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right), \\ \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{DE}\right] \geq 0, & \kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}. \end{cases}$$

Therefore, given monotonicity (in either direction) of the potential outcomes $Y_i(1,0)$ and $Y_i(0,0)$, the estimator $\widehat{DE}$ is attenuated *away* from the null of no direct effects if $\kappa > 1$ and attenuated *towards* the null if $\kappa \in \left(\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}, 1\right)$, and changes sign if $\kappa < \min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}$. Since π_i^e is typically small, then $\min_{i\in\mathcal{R}_e}\left\{1 - \frac{1}{\pi_i^e}\right\}$ is a large negative number, and it is unlikely that κ is smaller than that value in practice. Thus, in practice, we expect that the estimator $\widehat{DE}$ is attenuated away from the null if $\kappa > 1$ and towards the null if $\kappa < 1$ for most realistic scenarios.

B Extension to three-level exposure mapping

We extend the framework to a three-level exposure mapping for alters, $F_i \in \{0, 1, 2+\}$, indicating exposure to zero, exactly one, or two or more treated neighbors, respectively. We consider only the indirect effect estimand and estimators. We derive bias expressions for the naive HT estimator and propose an adjusted estimator to correct for the bias. The adjusted estimators relies on the same sensitivity parameter of alter–ego edges probabilties ρ^a_{ij} (9) defined in Section 3.3. However, we require an additional sensitivity parameter, similar to the interaction parameter κ for the bias-corrected direct effect estimator given in Section 3.2.

Let $S_i = \sum_{j \neq i} Z_j A_{ij}$ be the true number of treated neighbors of unit i in the population network $\boldsymbol{A}$. For alters $i \in \mathcal{R}_a$, we can write $S_i = \sum_{j \in \mathcal{R}_e} Z_j A_{ij}$. The modified exposure mapping (Assumption 2) is

Assumption B.1 (Three-Level Exposure Mapping)**.** *Let*

$$F(\boldsymbol{Z}_{-i}, \boldsymbol{A}) = \begin{cases} 0, & S_i = 0 \\ 1, & S_i = 1 \\ 2+, & S_i \geq 2 \end{cases}$$

For any $\boldsymbol{z}, \boldsymbol{z}' \in \mathcal{Z}$, if $z_i = z'_i$ and $F(\boldsymbol{z}_{-i}, \boldsymbol{A}) = F(\boldsymbol{z}'_{-i}, \boldsymbol{A})$, then $Y_i(\boldsymbol{z}) = Y_i(\boldsymbol{z}')$.

By Assumption B.1, we can write $Y_i(\boldsymbol{Z}) = Y_i(Z_i, F_i)$, where $F_i \in \{0, 1, 2+\}$. Thus, each unit now have six potential outcomes, instead of four. The modified consistency (Assumption 3) is

Assumption B.2 (Consistency (Three-Level))**.** $Y_i = \sum_{z \in \{0,1\}} \sum_{f \in \{0,1,2+\}} Y_i(0, f) \mathbb{I}\{Z_i = 0, F_i = f\}$.

The estimand of interest remains the sample average indirect effect of exposure to a single versus no treated neighbors (1)

$$IE = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} Y_i(0, 1) - Y_i(0, 0).$$

Bias of the naive estimator

The naive HT estimator $\widehat{IE}$ remains as defined in (3). Its expectation is $\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] = \frac{1}{n_a} \sum_{i \in \mathcal{R}_a} (E_{i,1} - E_{i,0})$, where

$$E_{i,1} = \mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\} Y_i}{p_z}\right]$$
$$E_{i,0} = \mathbb{E}_{\boldsymbol{Z}}\left[\frac{\mathbb{I}\{\widetilde{F}_i = 0\} Y_i}{1 - p_z}\right].$$

The issue is that observed exposures $\widetilde{F}_i = 1$ can corresponds to true exposures of either $F_i = 1$ or $F_i = 2+$, and observed exposures $\widetilde{F}_i = 0$ can corresponds to true exposures of $F_i = 0$, $F_i = 1$, or $F_i = 2+$. Therefore, the observed outcomes Y_i under observed exposures $\widetilde{F}_i$ are mixtures of potential outcomes under different true exposures F_i.

Define $S_{i,-e(i)} = \sum_{j \in \mathcal{R}_e^{e(i)}} Z_j A_{ij}$ as the number of treated neighbors of alter i *other than* its own ego $e(i)$. This represents the possible exposure from contamination between the ego-networks.

Expectation for $E_{i,1}$: If $\widetilde{F}_i = 1$, then $Z_{e(i)} = 1$. The true number of treated neighbors is $S_i = 1 + S_{i,-e(i)}$. Given that $Z_{e(i)} = 1$,

- The true exposure is $F_i = 1$ if $S_i = 1$, which implies $S_{i,-e(i)} = 0$.
- The true exposure is $F_i = 2+$ if $S_i \geq 2$, which implies $S_{i,-e(i)} \geq 1$.

By Assumptions B.1-B.2,

$$
\begin{aligned}
E_{i,1} &= \frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{\widetilde{F}_i = 1, F_i = 1\}Y_i(0,1) + \mathbb{I}\{\widetilde{F}_i = 1, F_i = 2+\}Y_i(0,2+)\right] \\
&= \frac{1}{p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\mathbb{I}\{Z_{e(i)} = 1, S_{i,-e(i)} = 0\}Y_i(0,1) + \mathbb{I}\{Z_{e(i)} = 1, S_{i,-e(i)} \geq 1\}Y_i(0,2+)\right] \\
&= \Pr(S_{i,-e(i)} = 0)Y_i(0,1) + \Pr(S_{i,-e(i)} \geq 1)Y_i(0,2+).
\end{aligned}
$$

The last step holds because $S_{i,-e(i)}$ is independent of $Z_{e(i)}$ under Assumption 1, and $\Pr(Z_{e(i)} = 1) = p_z$.

Expectation for $E_{i,0}$: If $\widetilde{F}_i = 0$, then $Z_{e(i)} = 0$. The true number of treated neighbors is $S_i = S_{i,-e(i)}$. Given that $Z_{e(i)} = 0$,

- The true exposure is $F_i = 0$ if $S_i = 0$, which implies $S_{i,-e(i)} = 0$.
- The true exposure is $F_i = 1$ if $S_i = 1$, which implies $S_{i,-e(i)} = 1$.
- The true exposure is $F_i = 2+$ if $S_i \geq 2$, which implies $S_{i,-e(i)} \geq 2$.

Therefore,

$$
\begin{aligned}
E_{i,0} &= \frac{1}{1-p_z}\mathbb{E}_{\boldsymbol{Z}}\left[\sum_{f\in\{0,1,2+\}} \mathbb{I}\{\widetilde{F}_i = 0, F_i = f\}Y_i(0,f)\right] \\
&= \Pr(S_{i,-e(i)} = 0)Y_i(0,0) + \Pr(S_{i,-e(i)} = 1)Y_i(0,1) + \Pr(S_{i,-e(i)} \geq 2)Y_i(0,2+).
\end{aligned}
$$

Exposures probabilities. As in Section 3.3, we assume conditional edge independence given ρ^a_{ij}, defined in (9). The random variable $S_{i,-e(i)}$ follows a Poisson-Binomial distribution with $n_e - 1$ parameters $\{p_z\rho^a_{ij}\}_{j\in\mathcal{R}_e^{e(i)}}$. Let $\pi^*_{i,k} = \Pr(S_{i,-e(i)} = k)$. The required probabilities are

$$
\begin{aligned}
\pi^*_{i,0} = \Pr(S_{i,-e(i)} = 0) &= \prod_{j\in\mathcal{R}_e^{e(i)}} (1 - p_z\rho^a_{ij}) \\
\pi^*_{i,1} = \Pr(S_{i,-e(i)} = 1) &= \sum_{j\in\mathcal{R}_e^{e(i)}} \left[p_z\rho^a_{ij} \prod_{l\in\mathcal{R}_e^{e(i)}\setminus\{j\}} (1 - p_z\rho^a_{il})\right] \\
\pi^*_{i,2+} = \Pr(S_{i,-e(i)} \geq 2) &= 1 - \pi^*_{i,0} - \pi^*_{i,1}.
\end{aligned}
$$

For large n_e, analytic computation of $\pi^*_{i,1}$ can be challenging, but approximation methods exist (Hong, 2013).

Bias expression. First, note that $\Pr(S_{i,-e(i)} \geq 1) = 1 - \Pr(S_{i,-e(i)} = 0) = 1 - \pi^*_{i,0}$. Substituting these probabilities, the expectations of the two components of the naive estimator are:

$$
\begin{aligned}
E_{i,1} &= \pi^*_{i,0}Y_i(0,1) + (1 - \pi^*_{i,0})Y_i(0,2+) \\
E_{i,0} &= \pi^*_{i,0}Y_i(0,0) + \pi^*_{i,1}Y_i(0,1) + (1 - \pi^*_{i,0} - \pi^*_{i,1})^*Y_i(0,2+)
\end{aligned}
$$

The expectation of the naive estimator is:

$$
\begin{aligned}
\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] &= \frac{1}{n_e}\sum_{i\in\mathcal{R}_a}(E_{i,1} - E_{i,0}) \\
&= \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\left[-\pi^*_{i,0}Y_i(0,0) + (\pi^*_{i,0} - \pi^*_{i,1})Y_i(0,1) + \pi^*_{i,1}Y_i(0,2+)\right] \qquad \text{(B.1)} \\
&= IE + \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\left[\pi^*_{i,1}\left\{Y_i(0,2+) - Y_i(0,1)\right\} - (1 - \pi^*_{i,0})\left\{Y_i(0,1) - Y_i(0,0)\right\}\right].
\end{aligned}
$$

Which implies that the naive estimator is also biased for the indirect effect estimand. The bias depends on the indirect effects of 2+ exposures relative to 1 exposures and of 1 exposures relative to 0 exposures, weighted by the exposure probabilities $\pi^*_{i,1}$ and $1 - \pi^*_{i,0}$, respectively.

Bias-corrected estimator

The resulting expected value (B.1) is a linear combination of the unknown potential outcomes $\{Y_i(0,0), Y_i(0,1), Y_i(0,2+)\}$. The problem is that the observed data do not provide enough information to identify all three potential outcomes for each alter $i \in \mathcal{R}_a$. To achieve identification, we introduce an additional sensitivity parameter δ via a structural assumption on the potential outcomes, analogous to Assumption 4.

Assumption B.3. *There exists a constant δ such that $Y_i(0,2+) - Y_i(0,0) = \delta \{Y_i(0,1) - Y_i(0,0)\}$ for all alters $i \in \mathcal{R}_a$.*

The sensitivity parameter δ in Assumption B.3 provides an intuitive way to model the dose-response relationship of alter exposure. It is best interpreted through the *marginal effect of additional exposure*. The effect of first exposure is $Y_i(0,1) - Y_i(0,0)$. The *additional* effect of the second-and-higher exposure is $Y_i(0,2+) - Y_i(0,1)$. Assummption B.3 implies a direct relationship

$$Y_i(0,2+) - Y_i(0,1) = (\delta - 1)\{Y_i(0,1) - Y_i(0,0)\}$$

Researchers can therefore specify plausible values for δ based on their domain knowledge.

- $\delta = 1$. The effect *saturates* after one exposure, and any further contamination is irrelevant. This recovers the binary exposure model.
- $\delta > 1$. There is a *reinforcing* cumulative dose effect, where a second exposure adds benefit. For example, $\delta = 2$ implies the same marginal effect between first and second-and-higher exposure. However, $\delta \in (1,2)$ implies that marginal effect between first and zero exposure is higher than the marginal effect between second-or-higher and first exposure.
- $\delta < 1$. Additional exposures *diminish* the effect of the first. For example, exposure to more than one treated neighbor may create confusion or mistrust, reducing the benefit of exposure.

This framework allows researchers to conduct sensitivity analysis by specifying a plausible range for the marginal effect of exposures, rather than the absolute value of $Y_i(0,2+)$ (e.g., by deriving sharp, yet informative, bounds). Rearranging the statement in Assumption B.3, we have

$$Y_i(0,2+) = (1-\delta)Y_i(0,0) + \delta Y_i(0,1). \tag{B.2}$$

Using (B.2), the expectation in (B.1) can be written as,

$$\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] = \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\left[\pi^*_{i,0} + \pi^*_{i,1}(\delta-1)\right]\left[Y_i(0,1) - Y_i(0,0)\right], \tag{B.3}$$

which implies the bias-corrected estimator of indirect effects under the three-level exposures

$$\widehat{IE}_{adj} = \frac{1}{n_a}\sum_{i\in\mathcal{R}_a}\frac{1}{\pi^*_{i,0} + \pi^*_{i,1}(\delta-1)}\left[\frac{\mathbb{I}\{\widetilde{F}_i = 1\}Y_i}{p_z} - \frac{\mathbb{I}\{\widetilde{F}_i = 0\}Y_i}{1-p_z}\right]. \tag{B.4}$$

The adjusted estimator $\widehat{IE}_{adj}$ is design-unbiased for the indirect effect estimand IE

$$\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}_{adj}\right] = IE.$$

Moreover, (B.3) implies that the naive estimator $\widehat{IE}$ is unbiased if

- $\pi^*_{i,0} = 1$. This corresponds to no contamination between ego-networks (no additional treated neighbors beyond the ego). That is equivalent to $\rho^a_{ij} = 0$ for all alter–ego edges.
- $\delta = 1 + \frac{1-\pi^*_{i,0}}{\pi^*_{i,1}} = 2 + \frac{\pi^*_{i,2+}}{\pi^*_{i,1}}$ for all $i \in \mathcal{R}_a$. This implies that the marginal effect of additional exposure is exactly offset by the contamination between ego-networks.

Relation to the two-level exposure model. Note that if we set $\delta = 1$ in (B.4), then $Y_i(0, 2+) = Y_i(0, 1)$, and the additional exposure level 2+ collapses into level 1. Recall that $\pi_i^a = \Pr(F_i = 1 \mid i \in \mathcal{R}_a)$ is the probability that alter i is exposed to *at least* one treated neighbor under the two-level exposure mapping. We can write

$$\begin{aligned}
1 - \pi_i^a &= \Pr(F_i = 0 \mid Z_{e(i)} = 1 \in \mathcal{R}_a) \Pr(Z_{e(i)} = 1 \mid i \in \mathcal{R}_a) \\
&\quad + \Pr(F_i = 0 \mid Z_{e(i)} = 0 \in \mathcal{R}_a) \Pr(Z_{e(i)} = 0 \mid i \in \mathcal{R}_a) \\
&= \Pr(S_{i,-e(i)} = 0)(1 - p_z) \\
&= \pi_{i,0}^*(1 - p_z).
\end{aligned}$$

That is, $\pi_{i,0}^* = \frac{1-\pi_i^a}{1-p_z}$. Substituting this into (B.4) with $\delta = 1$, we recover the bias-corrected estimator for the two-level exposure mapping given in (5). Therefore, the two-level exposure model can be viewed as a special case of the three-level exposure model with saturation of effects after the first exposure.

Attenuation and interplay between δ and $\pi_{i,k}^*$. If $Y_i(0, 1) - Y_i(0, 0) \geq 0$ for all alters $i \in \mathcal{R}_a$, then

$$\begin{cases}
\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \geq IE \geq 0, & \delta \geq \max_{i \in \mathcal{R}_a}\left\{2 + \frac{\pi_{i,2+}^*}{\pi_{i,1}^*}\right\}, \\
0 \leq \mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \leq IE, & \delta \in \left(\max_{i \in \mathcal{R}_a}\left\{2 - \frac{1-\pi_{i,2+}^*}{\pi_{i,1}^*}\right\}, \min_{i \in \mathcal{R}_a}\left\{2 + \frac{\pi_{i,2+}^*}{\pi_{i,1}^*}\right\}\right), \\
\mathbb{E}_{\boldsymbol{Z}}\left[\widehat{IE}\right] \leq 0 \leq IE, & \delta \leq \min_{i \in \mathcal{R}_a}\left\{2 - \frac{1-\pi_{i,2+}^*}{\pi_{i,1}^*}\right\}.
\end{cases}$$

Similarly, if $Y_i(0, 1) - Y_i(0, 0) \leq 0$ for all alters $i \in \mathcal{R}_a$, we obtain the opposite inequalities. Therefore, given monotonicity (in either direction) of the potential outcomes $Y_i(0, 1)$ and $Y_i(0, 0)$, the estimator $\widehat{IE}$ is either attenuated away, towards, or changes sign relative to the true indirect effect IE, depending on the value of the sensitivity parameter δ.

Similarly to the augmented bias-corrected estimator (6), we can define an augmented version of the three-level exposure bias-corrected estimator by adding outcome models for the conditional expectations $\mu_i^a(f) = \mathbb{E}\left[Y_i \mid \widetilde{F}_i = f, \boldsymbol{X}_i, i \in \mathcal{R}_a\right]$ for $f \in \{0, 1\}$. Variance estimation for $\widehat{IE}_{adj}$ can be done using the same approaches described in Section A for the bias-corrected indirect effects estimators under two-level exposures.

Applying the sensitivity analysis

The bias-corrected estimator (B.4) can be used in the same way as the bias-corrected estimators under two-level exposure mapping described in Section 3.4. That is, researchers can specify plausible ranges for the sensitivity parameters ρ_{ij}^a and δ based on their domain knowledge, and condust grid sensitivity analysis or probabilistic bias analysis. The main difference is the specification of the additional sensitivity parameter δ, and the a bit more complicated computation of the exposure probabilities $\pi_{i,0}^*$ and $\pi_{i,1}^*$. The latter can be easily computed in `R` using the approximations developed by Hong (2013) and are implemented in several packages.

C Probabilistic bias analysis

We give a short overview of the technical motivation behind probabilistic bias analysis (PBA). For a more detailed treatment, we refer the reader to Greenland (2005) and Fox (2021).

Assume we observe data $\boldsymbol{D}$ and are interested in estimating a causal effect θ. We have an estimator $\widehat{\theta}(\boldsymbol{D})$ based on the observed data, but we know that it is biased. Namely,

$$\mathbb{E}_{\boldsymbol{D}}\left[\widehat{\theta}(\boldsymbol{D})\right] \neq \theta.$$

Assume we can find a bias parameters ρ and construct an estimator $\widehat{\theta}_{adj}(\boldsymbol{D}, \rho)$ such that given ρ it is unbiased (w.r.t. to the sampling distribution of the data):

$$\mathbb{E}_{\boldsymbol{D}}\left[\widehat{\theta}_{adj}(\boldsymbol{D}, \rho) \mid \rho\right] = \theta.$$

However, the bias parameters ρ are typically unknown. In PBA, we specify a probability distribution for the bias parameters ρ based on prior knowledge or assumptions. This distribution reflects our uncertainty about the true value of the bias parameters. Let $p(\rho)$ denote this distribution.

With a slight abuse of notation, write $\widehat{\theta}_{adj}$ as random variable in the data $\boldsymbol{D}$ and the bias parameters ρ that represent the adjusted estimator given the data and bias parameters. Using this prior, we can derive a "posterior" distribution of the bias-corrected estimator $\widehat{\theta}_{adj}$ given the data $\boldsymbol{D}$ as follows.

$$p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}\right) = \int p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}, \rho\right) p(\rho) d\rho.$$

Which can be approximated through Monte Carlo integration by drawing M samples $\rho^{(1)}, \ldots, \rho^{(M)}$ from $p(\rho)$ and summarizing the distribution by, e.g., the average and percentiles. To account for the sampling variability as well, note that

$$p\left(\widehat{\theta}_{adj}\right) = \int_{\boldsymbol{D}} \int_{\rho} p\left(\widehat{\theta}_{adj} \mid \boldsymbol{D}, \rho\right) p(\rho) p(\boldsymbol{D}) d\rho d\boldsymbol{D},$$

where $p(\boldsymbol{D})$ is the sampling distribution of the data. Assume we the distribution of $\widehat{\theta}_{adj}(\boldsymbol{D}, \rho^{(m)})$ with normal distribution. Then for each draw $\rho^{(m)}$ from $p(\rho)$, estimate the adjusted estimator and add random noise according to the sampling distribution to obtain a draw from $p\left(\widehat{\theta}_{adj}\right)$. Alternatively, it is possible use a two-dimensional bootstrap that, in each iteration, sample a sensitivity parameter, draw a bootstrap sample of the data, and compute the adjusted estimator in the bootstrap sample. Repeating this procedure many times allows us to approximate this integral.

In our case, the only randomness in the data is due to the randomization $\boldsymbol{Z}$. Thus, we set $\widehat{\theta}(\boldsymbol{Z}) = \widehat{IE}$ or $\widehat{DE}$, the bias parameters $\rho = \pi_i^a$ or π_i^e and κ, and $\widehat{\theta}_{adj}(\boldsymbol{Z}, \rho) = \widehat{IE}_{adj}$ or $\widehat{DE}_{adj}$. Which motivates the PBA procedure described in Section 3.4.

D Simulations

We generated three covariates for each unit.

$$\begin{aligned} X_{i1} &\sim \text{Bernoulli}(p_1), \\ X_{i2} &\sim \text{Bernoulli}(p_2), \\ X_{i3} &\sim \text{Normal}(0, 2^2), \end{aligned}$$

where $p_1 = 0.6, p_2 = 0.2$ for egos and $p_1 = 0.5, p_2 = 0.3$ for alters. The covariates coefficents in the potential outcome models were set to $\beta_e = (-0.5, -0.3, 0.2)$ for egos and $\beta_a = (-0.4, -0.2, 0.1)$ for alters. Heterogeneous edges probabilities were generated with weights computed using the Euclidean distance $\|\cdot\|_2$ between the covariates of units. The nonparameteric ego-network bootstrap was done by resampling egos-networks with replacement $B = 5 \times 10^3$ times.

Table D.1 reports the simulation results for direct effects with heterogeneous contamination and augmented estimators, which complement the results presented in the main text. Tables D.2 and D.3 report the simulation results for indirect and direct effects, respectively, with heterogeneous exposure probabilities and non-augmented estimators. Tables D.4 and D.5 report the simulation results for indirect and direct effects, respectively, with homogeneous exposure probabilities and augmented estimators. All results are similar to those presented in the main text.

Table D.1: Simulation results for direct effects with heterogeneous exposure probabilities and augmented estimators.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_e = 50, DE = 0.345$	Heterogeneous	Analytic	0.013	0.966	0.860
		Bootstrap	0.013	0.964	0.864
	Homogeneous	Analytic	0.010	0.966	0.869
		Bootstrap	0.010	0.967	0.864
	Naive	Analytic	0.089	0.684	0.866
		Bootstrap	0.089	0.687	0.867
$m_e = 150, DE = 0.355$	Heterogeneous	Analytic	0.016	0.936	0.930
		Bootstrap	0.016	0.936	0.928
	Homogeneous	Analytic	0.015	0.939	0.930
		Bootstrap	0.015	0.934	0.932
	Naive	Analytic	0.211	0.044	0.929
		Bootstrap	0.210	0.046	0.935
$m_e = 250, DE = 0.345$	Heterogeneous	Analytic	0.014	0.959	0.823
		Bootstrap	0.014	0.956	0.832
	Homogeneous	Analytic	0.014	0.957	0.822
		Bootstrap	0.014	0.956	0.818
	Naive	Analytic	0.270	0.001	0.820
		Bootstrap	0.270	0.001	0.820

Table D.2: Simulation results for indirect effects with heterogeneous exposure probabilities and non-augmented estimators.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_a = 100, IE = 0.358$	Heterogeneous	Analytic	0.016	0.972	0.855
		Bootstrap	0.016	0.972	0.857
	Homogeneous	Analytic	0.015	0.973	0.855
		Bootstrap	0.015	0.972	0.857
	Naive	Analytic	-0.029	0.943	0.855
		Bootstrap	-0.029	0.943	0.857
$m_a = 200, IE = 0.335$	Heterogeneous	Analytic	0.006	0.974	0.872
		Bootstrap	0.006	0.972	0.874
	Homogeneous	Analytic	0.008	0.975	0.867
		Bootstrap	0.008	0.974	0.869
	Naive	Analytic	-0.068	0.831	0.867
		Bootstrap	-0.068	0.829	0.869
$m_a = 300, IE = 0.318$	Heterogeneous	Analytic	-0.003	0.975	0.871
		Bootstrap	-0.003	0.975	0.873
	Homogeneous	Analytic	0.003	0.975	0.872
		Bootstrap	0.003	0.975	0.874
	Naive	Analytic	-0.097	0.697	0.872
		Bootstrap	-0.097	0.696	0.874

Table D.3: Simulation results for direct effects with heterogeneous exposure probabilities and non-augmented estimators.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_e = 50, DE = 0.345$	Heterogeneous	Analytic	0.014	0.950	0.968
		Bootstrap	0.014	0.950	0.970
	Homogeneous	Analytic	0.011	0.950	0.972
		Bootstrap	0.011	0.949	0.975
	Naive	Analytic	0.090	0.776	0.972
		Bootstrap	0.090	0.773	0.975
$m_e = 150, DE = 0.355$	Heterogeneous	Analytic	0.017	0.932	1.035
		Bootstrap	0.017	0.931	1.038
	Homogeneous	Analytic	0.016	0.932	1.042
		Bootstrap	0.016	0.933	1.045
	Naive	Analytic	0.212	0.249	1.042
		Bootstrap	0.212	0.250	1.045
$m_e = 250, DE = 0.345$	Heterogeneous	Analytic	0.015	0.944	0.985
		Bootstrap	0.015	0.942	0.988
	Homogeneous	Analytic	0.014	0.940	0.988
		Bootstrap	0.014	0.940	0.991
	Naive	Analytic	0.271	0.109	0.988
		Bootstrap	0.271	0.108	0.991

Table D.4: Simulation results for indirect effects with homogeneous exposure probabilities and augmented estimators.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_a = 100, IE = 0.362$	Heterogeneous	Analytic	0.010	0.987	0.741
		Bootstrap	0.010	0.988	0.740
	Homogeneous	Analytic	0.009	0.988	0.742
		Bootstrap	0.009	0.987	0.745
	Naive	Analytic	-0.035	0.943	0.743
		Bootstrap	-0.035	0.942	0.741
$m_a = 200, IE = 0.32$	Heterogeneous	Analytic	-0.002	0.976	0.867
		Bootstrap	-0.002	0.975	0.870
	Homogeneous	Analytic	-0.001	0.977	0.865
		Bootstrap	-0.001	0.976	0.864
	Naive	Analytic	-0.072	0.681	0.864
		Bootstrap	-0.071	0.678	0.864
$m_a = 300, IE = 0.372$	Heterogeneous	Analytic	-0.039	0.949	0.860
		Bootstrap	-0.039	0.949	0.861
	Homogeneous	Analytic	-0.046	0.931	0.861
		Bootstrap	-0.046	0.931	0.863
	Naive	Analytic	-0.148	0.076	0.864
		Bootstrap	-0.148	0.075	0.864

Table D.5: Simulation results for direct effects with homogeneous exposure probabilities and augmented estimators.

Scenario	Specification	Variance	Error	Coverage	SD/SE
$m_e = 50, DE = 0.32$	Heterogeneous	Analytic	0.021	0.953	0.887
		Bootstrap	0.021	0.954	0.887
	Homogeneous	Analytic	0.019	0.955	0.882
		Bootstrap	0.019	0.955	0.885
	Naive	Analytic	0.094	0.666	0.884
		Bootstrap	0.094	0.664	0.885
$m_e = 150, DE = 0.295$	Heterogeneous	Analytic	0.093	0.297	0.943
		Bootstrap	0.094	0.303	0.950
	Homogeneous	Analytic	0.088	0.333	0.957
		Bootstrap	0.088	0.332	0.958
	Naive	Analytic	0.290	0.002	0.956
		Bootstrap	0.290	0.001	0.957
$m_e = 250, DE = 0.38$	Heterogeneous	Analytic	-0.027	0.913	0.918
		Bootstrap	-0.027	0.910	0.920
	Homogeneous	Analytic	-0.032	0.873	0.911
		Bootstrap	-0.032	0.877	0.912
	Naive	Analytic	0.216	0.024	0.909
		Bootstrap	0.217	0.026	0.909

E Data analysis

Figure E.1 shows the distribution of number of alters per ego-network in the data. The average and median number is 2, the range is from 1 to 6.

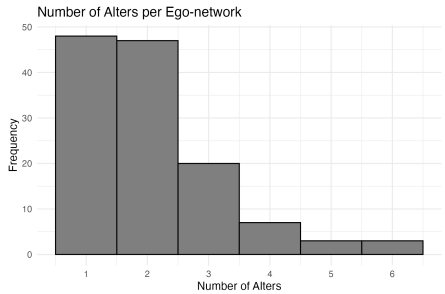

Figure E.1: Distribution of number of alters per ego-network.

Figure E.2 shows the results of the grid sensitivity analysis (Section 5.1) for the direct effect given that $\kappa = 2$. This presenation, similar to that of indirect effects, enables us to see how changing the expected number of missing ego–ego edges impact the estimated direct effects. Clearly, for both specifications, the estiamted effects approach zero as the contamination level increases. However, all confidence intervals include zero for all levels of contamination, indicating that the direct effect is not statistically significant even without accounting for contamination.

Figure E.3 shows the results of the grid sensitivity analysis for indirect effects using the augmented estimator. The structure is similar to that of the non-augmented estimator presented in the main text. However, the augmented estimator had values closer to zero for all levels of contamination (including no contamination).

Figure E.4 shows the results of the probabilistic bias analysis for direct effects using the augmented estimator. The results are similar to those of the non-augmented estimator presented in the main text.

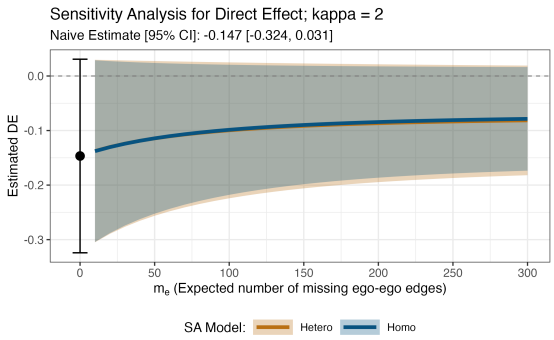

Figure E.2: Grid sensitivity analysis for direct effects with $\kappa = 2$. Non-augmented estimator.

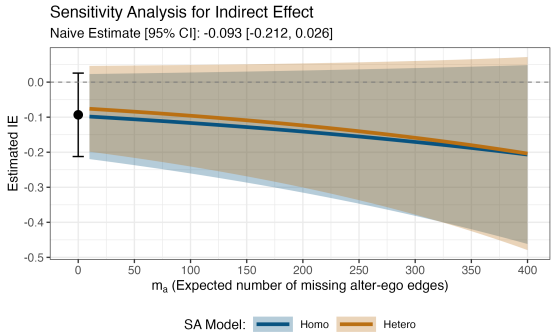

Figure E.3: Grid sensitivity analysis for indirect effects with augmented estimator.

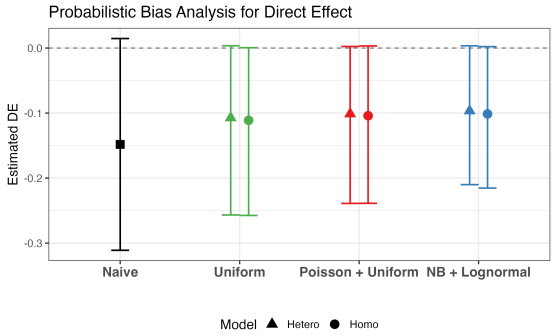

Figure E.4: Probabilistic bias analysis for direct effects with augmented estimator.